

Generic Diagnostic Vehicle Specification

VFDSVAEEP0001-V1.6

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Specification Book

Scope

Applicable Document of Part 1

Common Electrical Specification for Supplier
Generic Diagnostic Specification for all Vehicle programs.

Range of Validity

Vinfast

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Approval Confirmed By

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1 General Information

1.1 Language

This specification is written in English.

1.2 Change History

Version	Chapter	Brief Description	Date	Responsible person
1.0	all	Document creation	2021-09-20	Ashad Qureshi Daniel Canelopol
1.1	4.1	Remove table 10	2021-11-15	Ashad Qureshi
1.2	3.2	Update speed signal name Update section 3.2 update extended data with record #12 Section 2.0 update with statement.	2022-01-05	Ashad Qureshi
1.3	6.6.2 Section 3.6, 3.8 3.14 1.2 3.2	Change of soft reset requirement from condition C to M and extended snapshot changed to Record changed sentence. Detail of extended record number clarified. LIN specification added Update the requirement of diagnostic preconditions for general diagnostics. Table numbers update and updated table for DTC extended data	2022-01-14	Ashad Qureshi
1.4	3.2	Changed obsolete vehicle speed and speed valid signals table4 Added a new VIN to cater for Case 2 and text amended	2022-02-08	Ashad Qureshi
	3.30		17/03/2022	Ashad Qureshi
1.4	3.2	Wording changed and paragraph clarified	08/04/22	Ashad Qureshi
	3.2	Vehicle speed status corrected table 5 and removed the motor (and E-motor OFF for BEV)		
	3.3	Wait time post ignition increased from 1S to 1.5S and reworded the paragraph and network logging DTC voltage requirements defined		
	6.3	ODST test bit 0 definition clarified Fig 6 replaced VOD reference document defined VOD section clarified		
	3.5	table 2 updated with ISO document		
	3.18	DTC mature de-mature strategy clarified. Title heading number corrected section 6 to 6.35		
	1.3	Table 33 amended to make some sub-functions of service 19 mandatory Ref of ISO14229 added Sentence clarified		
	6.0	Requirement clarified		

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1.5	6.26	Requirement clarified	11/07/2022	Ashad Qureshi Ashad Qureshi
	3.14	Requirement clarified		
	2.2	Requirement clarified		
	2.3	Requirement clarified		
	3.4	Requirement clarified		
	3.5	Requirement clarified		
	3.6	Requirement clarified		
	3.8	Requirement clarified		
	3.11	Table updated with timing		
	3.17	Requirement clarified		
	3.21	Requirement clarified		
	3.27	Note updated		
	4.3	Requirement clarified		
	4.5	Requirement clarified		
	4.6	Requirement clarified		
	6.32	Requirement clarified		
	4.8	Requirement clarified		
	3.20	Requirement clarified		
	5.3	Table 20 updated		
	6.2	Table 23 updated		
	6.5	Requirement Updated		
	6.6	Tables 17, 18 updated		
	6.7	ECU reset service requirement clarified		
	6.12	Service 29 requirement detailed		
	6.15	Table 4,5 updated		
	6.28	S3 client and server clarified		
	6.8	Functional addressing clarified		
	6.13	Table 9.1 updated		
	6.31	Requirement clarified		
	3.2	Requirement clarified		
	4.3	Added new section 3.14 and table 9.1 moved under section 3.14		
	4.6			
	3.7	Table 2 updated		
	3.13	Requirement clarified and synced with CAN DB		
	3.8,3.9,3.10	DD01, DD02		
	3.14	Requirement updated		
	1.3	Table 16 updated		
	3.6	DTC Strategy defined		
	4.9	Table 16 updated to include P4 CAN time and St_min		
	4.3	Table 4 updated		
	3.4	Requirement clarified		
	4.3			
	3.2	Requirement clarified		
	3.2			
6.13	Types of ECU resets defined			
3.6	Global and local snapshots definition clarified			
1.3&4.3	ISO Reference amended			
6.31	Service 29 flow added			
3.2	Condition for diagnostics clarified			
	Definition clarified by adding ref			

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1.6	3.4	document ODST text clarified	08/02/2023	Ashad Qureshi
	3.5	Tables 9 and 9.1 record type removed from the title to stop confusion with snapshot.	15/02/2023	Ashad Qureshi
	3.7-3.14	Paragraph corrected.	15/02/2023	Ashad Qureshi
	3.3	3.6 paragraph corrected.	21/02/2023	Ashad Qureshi
	6.16	Definition for supporting TP clarified Table updated	02/02/2023	Ashad Qureshi
	6.7	3.13 paragraph clarified	02/02/2023	Ashad Qureshi
	3.10	Missing word added	"	"
	13.15	DTC clearing updated note on DTC clearing – "Hard Fault – Permanent DTC		
	4.3	ISO reference document corrected	09/03/2023	Ashad Qureshi
	3.13.1	New heading added for extended data		
	3.5	Corrected ODST and added table 5A	10/03/2023	Ashad Qureshi
	All	Grammar corrections	13/03/2023	
	3.21	Sentence corrected.	14/03/2023	Ashad Qureshi
	3.2	Paragraph clarified for condition as well as flash programming.	16/03/2023	Ashad Qureshi
	2.3		16/03/2023	Ashad Qureshi
	3.6	Sentence, corrected.	"	"
	6.5	ISO section defined.	"	"
	1.1	P2* and session handling clarified.	"	"
	3.11	Table 2 ref ISO27145-3	17/03/2023	Ashad Qureshi
	4.6	Definition for DTC severity added.	"	"
	6.24	Definition clarified	20/03/2023	"
	6.26	Table 33 updated	"	"
	6.3	Running of routines clarified.	"	"
	7.1	Paragraph clarified ref sessions.	"	"
	3.21/9	Paragraph clarified.	"	"
		Ref to Transport mode removed(not required by VinFast LV battery disconnected)	21/03/23	Steve Clarke
			"	"
	Document	Standardised 0xnn and deleted \$nn references to numbers plus other minor comprehension changes.	"	"
	3.32	Details added from ISO142229-1 and ISO 27145-3	"	"
			"	"
	6.6	Coding Session added P2*Server reference.	"	"
	3.11	Clarified DTC Severity	27/3/2023	Ashad Qureshi
	3.13	Added DTC Priority Definition and associated Note.	29/3/2023	Steve Clarke
	All	Headings renumbered	"	"

Table 1: Change History

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1.3 Reference Documents

Reference Number	Source	Title	Version or date	Document Number
1)	ISO	Road vehicles – Diagnostics systems – Diagnostic services	2020	ISO/DIS 14229-1-7
2)	ISO	Road vehicles — Communication between vehicle and external equipment for emissions-related diagnostics — Part 7: Data link security	2013	ISO15031-7
3)	ISO	Road vehicles — Communication between vehicle and external equipment for emissions-related diagnostics-part 6	2015	ISO15031-6
4)	ISO	Road vehicles — Communication between vehicle and external equipment for emissions-related diagnostics — Part 5: Emissions-related diagnostic services	2015	ISO 15031-5
5)	ISO	Road vehicles — Communication between vehicle and external equipment for emissions-related diagnostics — Part 4: External test equipment	2014	ISO 15031-4
6)	ISO	Road vehicles — Diagnostics on Controller Area Networks (CAN)	2021	ISO15765-4
7)	ISO	Road vehicles — Diagnostic communication over Controller Area Network (DoCAN) — Part 5: Specification for an in-vehicle network connected to the diagnostic link connector	2021	ISO15765-5
8)	ISO	Road vehicles — Diagnostic communication over Controller Area Network (DoCAN) — Part 2: Transport protocol and network layer services	2016	ISO15765-2
9)	ISO	Road vehicles — Local Interconnect Network LIN	2016	ISO17987-2
10)	ISO	Road vehicles — Modular vehicle communication interface	2017	ISO22900
11)	ISO	Road vehicles — Open diagnostic data exchange (ODX)	2018	ISO22901
12)	SAE	E/E Diagnostic Test Modes	2015	SAE J1979
13)	SAE	Diagnostic Trouble Code Definitions	2016	SAE J2012
14)	VF	VIN structure	2021	VF-HOMO-DOC-02
15)	VF	VIN Decipher	2021	
16)	CES	CES_Basic_Services_DID_PID_Definition	Latest	VFDSAAEEP0033
17)	CES	CES_Flash_Specification	Latest	CES_VINFAST_EE_VFDSXVEEP0035
18)	CES	CES_Coding_Specification	Latest	See note* VFDSP50000017 (VFDSVAEEP0004)
19)	CES	CES_ECU_Identifier_Specification	Latest	VFDSP50000013
20)	CES	CES_Security_Access_Algorithm	Latest	CES_VINFAST_EE_VFDS P50000020
21)	Gov	49 CFR 565 - VEHICLE IDENTIFICATION NUMBER (VIN) REQUIREMENTS*	Check	CFR part 565*
22)	VF	System_Schematics	Latest	
23)	AUTOSAR	Requirements on Diagnostics	Latest	AUTOSAR FO R19-11
24)	VF	CES_ECU LIN_Specification	Latest	CES_VINFAST_EE_VFDS P50000012
25)	ISO	Road vehicles — Environmental conditions and testing for electrical and electronic equipment	Latest	ISO 16750-2:2010(E)
26)	ISO	Road vehicles — Implementation of World-Wide Harmonized On-Board Diagnostics (WWH-OBD) communication	Latest	ISO27145-3
27)	ISO	Road vehicles — Controller area network (CAN) — Part 1: Data link layer and physical signalling	Latest	ISO11898

Table 2 requirements

* Note this VOD is for the current vehicle VFE 8,9.

The Standard's incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. It is the responsibility of the supplier to check for the latest publications where possible for the

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development of the ECU if there is uncertainty, then clarifications should be sought from Vinfast or their approved agents acting on their behalf as to which version of the requirement should be used.

*Obtainable from: U.S. Government Publishing Office <https://www.gpo.gov/fdsys/granule/CFR-2011-title49-vol6/CFR-2011-title49-vol6-part565/content-detail.html>

1.4 Document Conventions

In this document, the following terminology applies:

- ▶ “Shall” expresses an obligatory/mandatory requirement.
- ▶ “Should” expresses a recommendation or an advice.
- ▶ “Must” expresses a legal or normative requirement.
- ▶ “Will” expresses a precautionary consideration or an additional/optional feature.
- ▶ “May” expresses a permitted practice/method, not to be considered as a requirement.

2 Introduction

Though an attempt was made to cover all aspects of implementing VF requirements there may still be design attributes that may lack detail or are overlooked. As these and other attributes are further studied, requirements will be added or modified to reflect the latest industry and corporate directions and trends.

2.1 Purpose of Diagnostics

The purpose of the diagnostics can be defined by the following statements: -

- To validate correct assembly
- Check for presence of faults during service or repairs.
- To check for communication between the various network nodes that the ECU is connected or communicates through.
- To detect and store a fault regardless of the Off Board Tester making a diagnostic request to obtain this information.
- To report current failures when the On Demand Self-Test is run.
- Legal considerations for diagnostics any Software download solutions that are not via the usual diagnostic communications between the tester and the vehicle shall be reviewed for compliance to right to repair legislation i.e., cannot use proprietary methods to diagnose an ECU.

2.2 Scope

This document covers the basic requirements for Diagnostic task in the current project and describes the main Diagnostic requirements.

The documents:

- Basic DID Definition: - System are mandatory DIDs which need to be supported by all ECU suppliers, case by case basis as defined in the DID specification.
- Coding Specification: - Defines the requirement if the ECU needs to be coded for adapting ECU behaviour in a vehicle. While applying its general functionality via flash programming
- Flash Specification: - Defines/captures the requirement if the ECU needs to be flashed during production/EOL/After-sales activity.
- Security Access Algorithm: - Defines the requirement to implement security for services and

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functions which need to be protected.

follow the requirements of this specification.

The functionality should be based on the requirements of the Production and Service. Following types of diagnostic operations are described in this specification:

- Basic Diagnostic functionalities
- Basic ON-Board diagnostic functionality
- Basic OFF-Board diagnostic requirements
- Additional diagnostic features
- Linked specifications and documentations.

2.3 Terms and Definitions

OFF-Board communication includes all processes of data communication between a car and the tester device. The **OFF-Board diagnostic** communication is subject to special regulations of OBDII ISO 15031 and, SAE J1979 / SAE J17979-DA. All processes of this communication must be conforming to the requirements of Unified Diagnostic Service (UDS, ISO 14229).

Communication over K-Line with OBDII/EOBD relevant System are not allowed in Europe and North America since 2008. This communication must be done over Diagnostic CAN regarding ISO 15031 and ISO 11898.

On-Board communication describes the communication processes between the ECUs over the car communication network.

This automatically includes the **On-Board diagnosis** as an important point which allows self-diagnosis of electronic components and systems. That means that ECUs must be able to detect faults within themselves. On-Board diagnosis makes sure the information and functionality for the Off- Board diagnostic.

A special part of the **On-Board diagnosis** is the fault detection of emissions-related parts regarding OBDII/EOBD. Here the special requirements of the ISO 15031 (part 1 ... part 6) are binding and the requirements of the SAE J1979 / SAE J17979-DA shall be valid.

Note that this part of diagnostic content must be defined by ECU engineer and ECU supplier together, and OEM diagnostic engineer shall be responsible for checking and reviewing, integration testing engineer shall be responsible for validation.

On-Board Diagnostic is required to runs in three levels:

- Detection of error/fault
- Conditioning capturing the conditions surrounding the error/fault.
- Storing the fault and the data captured during the event in the form of snapshot.

Once a fault is detected it shall be assigned with a unique DTC with the relevant information about the fault must be stored within the ECU to be extracted by the Off-Board Tester during subsequent repairs for analysis and to help in determining the root cause of the fault. The emphasis is therefore on all suppliers to capture as much data as they possibly can during failure event as trying to capture the details surround the event may not be possible after the event.

Figure 1: Areas of On and OFF Board

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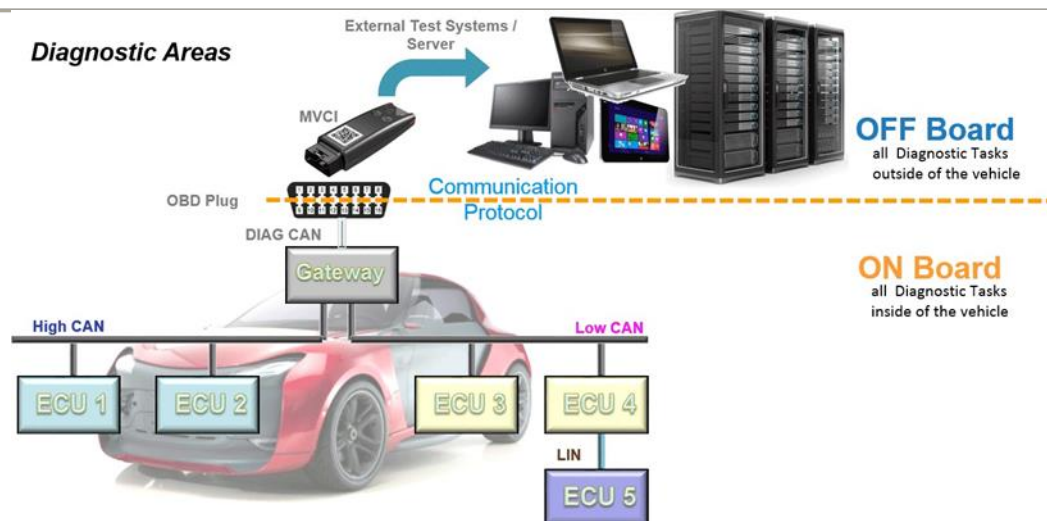


Fig:1

2.4 Abbreviations

DTC	Diagnostic Trouble Code
DID	Data Identifier
PID	Parameter Identifier
RID	Routine Identifier
FC	Flow Control
CAN	Control Area Network
EOBD	European On Board Diagnostic
ISO	International Organization for Standardisation
OBDII	On Board Diagnostic
SAE	Society of Automotive Engineers
KWP	Key Word Protocol
UDS	Unified Diagnostic Services
ODX	Open Diagnostic Data Exchange
SAE	Society of Automotive Engineers
M	Mandatory
U	User conditional
C	Conditional
O	Optional depending on ECU requirement
VIN	Vehicle Identification Number
DOIP	Diagnostic Over Internet Protocol
MFG	Manufacturing
OTA	Over The Air

Table 3: Abbreviations

3 Diagnostic Requirements Overview

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3.1 Basic Requirements

The ECUs must fulfil the requirements of the ISO 14229 UDS. A Gateway routes all necessary diagnostic commands and messages to the connected network components and has to provide the connection and communication to an OFF-Board Tester. As diagnostic data exchange format is a complete **ECU ODX template** for every phase shall be delivered to OEM from the suppliers of the ECUs.

3.2 Conditions for Diagnosis

For Diagnostics task on the vehicle, vehicle safety and user safety shall be considered. Therefore, Diagnostic Services required to be supported where ECU either needs software update or where the ECU is requested to carry out a diagnostic task which is behind security whilst the vehicle is in motion, in this case some services will not be allowed, when such a request is made the ECU shall check the prerequisite conditions and if these are not met the ECU shall respond back with NRC0x22 (Conditions Not Correct). The prerequisites conditions are ignition on Stat terminal driver present and vehicle speed is <3km/h. These preconditions checks can only be checked whilst the ECU is in application mode with valid application. It is also expected for the ECU to check these preconditions autonomously/concurrently in background i.e., if the ECU has valid application and is requested to carry out a task which requires a precondition check then ECU will look at the signals as defined in table 4 and if the preconditions are satisfied then the ECU will carry out the task without any further checks being requested. If the ECU is in bootloader mode or is without a valid application, it is therefore not expected for that ECU to check the fulfilment of these prerequisites. For the details on which services are allowed to respond please see table 4.

When the vehicle is being conditioned for flash and as part of this process involves the disablement of all non-diagnostic communication messages using service 0x28, after this service is activated, all ECUs are required not to check the precondition, this state will remain for the duration of service 28 being active.

Table 4

No.	Diagnostic Preconditions	Message ID	CAN Signal	Signal Value	BEV	ICE
1	Ignition ON	0x112	STAT_Terminal	0x02 ON (Driver Present) or 0x04 Software Update OTA	X	X
2	Vehicle Not Moving	0x20D	VehicleSpd and Vehicle SpdSts	Less than 3km/h 0x00 Valid	X	X
Services which are allowed/not allowed to respond when vehicle is not stationary these conditions are to be monitored when exercising these services						
Service	Vehicle stationary IGN ON	Vehicle not Stationary IGN ON	Comments			
10 Diagnostic Session control	Allowed	*Not allowed 02/82/03/83	*Note Session 10 40/50 only allowed if it is required to have the vehicle running			
11 ECU Reset	Allowed	Not allowed				
14 Clear DTC	Allowed	Not allowed				
19 Read DTC	Allowed	Allowed				
22 Read Data By Identifier	Allowed	Allowed				
2E Write Data By Identifier	Allowed	Not allowed				
2F Input Output Control By Identifier	Allowed	Not allowed				
27 Security Access	Allowed	* Conditional	*Only allowed in session 10 40 if the vehicle needs to unlock the ECU for running routines			
28 Communication Control	*Allowed	Not allowed	*When service 28 disabled communication is disabled no precondition check should be carried out			
31 Routine Control	Allowed	* Conditional	* only allowed In session 10 40 if it is required to have the vehicle running			
34 Request Download	Allowed	Not allowed				
36 Transfer Data	Allowed	Not allowed				
37 Request Transfer Exit	Allowed	Not allowed				
3E Tester Present	Allowed	Allowed				

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85 Control DTC Setting	Allowed	Not allowed	
29 Authentication	*Allowed	Not allowed	*Allowed if the service is supported
2A Read Data By Periodic Identifier	*Allowed	Not allowed	*Allowed if the service is supported
2C Dynamically Define Data Identifier	*Allowed	Not allowed	*Allowed if the service is supported
38 File Transfer	*Allowed	Not allowed	*Allowed if the service is supported
83 Access Timing Parameter	*Allowed	Not allowed	*Allowed if the service is supported
84 Secured Data Transmission	*Allowed	Not allowed	*Allowed if the service is supported
86 Response On Event	*Allowed	Not allowed	*Allowed if the service is supported
87 Link Control	*Allowed	Not allowed	*Allowed if the service is supported
24 Read Scaling Data By Identifier	*Allowed	Not allowed	*Allowed if the service is supported

Table 4: Diagnostic precondition

Note: - If an ECU requires the vehicle to be running to calibrate a sensor for example, then the supplier shall support a special diagnostic session for this activity (i.e., 0x10 40/50). These additional sessions are required for when the ECU is fitted to the vehicle, when testing a standalone ECU these additional service sessions are not required.

If deviation from the above or any other conditions for which Vinfast requires conformity (e.g., for Production/Aftersales), the necessary conditions for diagnosis it shall be agreed and documented with the authorised Engineering/ Diagnostic team. Ensuring that correct process is followed, the deviation shall be authorised by designated team members whose names and contact details will be listed in this document or identified in the deviation process. The deviation must be officially listed and captured on a Vinfast single issue list, if this is required on a permanent basis there may be potential downstream customers who may be affected by this change and will need to be informed, else it shall be corrected by the supplier at the earliest opportunity in the program. The reference of the deviation will also be documented in the ODX.

For Flash/Programming tasks and Coding:

Before commencing an on vehicle flash the precondition check must be met.

All ECU's which support flash programming/coding shall check precondition and report back to the tester what has been requested.

These precondition checks are to be reported by the ECU in RID0x0203 response (CheckProgrammingPreConditions), prior to flash programming execution.

Based on result of RID0x0203 response from the ECU, the diagnostic tester tool will abort the flash programming session for the vehicle. The precondition shall also include reading of the HV via service 22 (DID 016C please see flash specification section 10 onwards).

Basically, a reprogramming or coding process shall only be allowed on a vehicle when the following preconditions are met: It is responsibility of the Tool operator/end user to ensure that vehicle is in correct state to meet this requirement. The ECU suppliers are not allowed to link the success or failure of the precondition to determine an ECU flash failure.

No.	Flashing Preconditions	Message ID	CAN Signal	Signal Value	BEV	ICE
1	Ignition ON (and E-motor OFF for BEV)	0x112	STAT_Terminal	0x02 ON (Driver Present) or 0x04 Software Update OTA	X	X
2	Vehicle Not Moving	0x20D	VehicleSpd and VehicleSpdSts	Less than 3km/h 0x00 Valid	X	X
3	System Voltage Ok	0x10A	*Ubatt	11.5 ~ 16v*	X	X

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4	Any other condition which can have a negative impact on electronic components or general functionality regarding vehicle safety i.e. HV off DID 0x016C						
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Table 5: Precondition check

*These are min and max voltage checks prior to commencing flashing, the ECU shall reject the flash request with conditions not correct if the voltage falls below or above the required threshold.

3.3 Diagnostic Trouble Codes

The ECU shall only log DTCs when Driver Present 02 status (0x112) is on and wait 1.5s post ignition as a prerequisite to ensure that vehicle along with all its other ECUs are under the same normal operating conditions.

Network DTCs shall not be logged below, or above voltage as defined by ISO 16750-2 E Ref (25).

The ECUs must detect fault conditions internally and store Diagnostic Trouble Codes (DTC), if the fault condition is reached (Self-diagnosis). Therefore, internally the ECUs have to check, e.g., the input and output for power supply, or the different signals and the CAN network.

If required, in addition the above ECU also have to detect fault conditions which are on the peripheral devices connected to it, i.e., Sensors, Actuators, etc. for which the ECU shall be responsible for.

DTCs are defined by the requirements of the OEM and suppliers working together with input from Development department, Production department and Aftersales department. Any opportunities to correct or enhance diagnostics contents, the supplier has the responsibility of leading this activity which includes to check, review, and manage DTC information and its definition.

3.4 Detection of Faults.

- The ECU should be able to identify all its faults for all components connected on each of its I/O, and external circuits connected to the ECU. The components of the external electrical circuits shall include cables. Please see ISO14229 Table 23 for services allowed in default and non-default sessions.

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- An ECU shall also be able to detect faults other than those in the external circuits connected to the ECU (e.g., all internal faults such as memory failure etc. in the ECU) any failure of the internal component which is not replaceable in service shall be deemed as ECU internal fault and a single DTC for all ECU internal failure type shall be flagged. It may be possible to distinguish different kind of ECU internal failure by associating a different failure byte for it. In the event it is the responsibility of the ECU supplier to capture any the data internally to the ECU surrounding the event which leads/causes the ECU to declare ECU internal fault. It is therefore, advisable for the ECU supplier to carry out any FMEA to identify any externally caused events which may lead to an ECU falsely declaring an ECU internal failure so that these could be identified and corrected by designing them out in the software to prevent from occurring.
- Connectors, sensors, and actuators (including both the electrical and the mechanical part of the sensors and actuators). A feasibility study is required, and the fault detection shall be implemented wherever a technical solution is available, and its cost is justified. The ECU supplier is responsible for the feasibility study. The feasibility study shall be approved by Vinfast or its approved agent of authority acting on their behalf. All detected faults shall be fully specified and documented by the supplier in the ODX with relevant DTCs and help text to assist service individual on the root cause and repair procedures.
- The smallest component replaceable in the workshop shall be identified by one unique DTC. Where possible, each failing part of the component should also be identified by a unique DTC so long as this enhances either the fault tracing capability and/or quality tracking/following up. Example short to ground, short to battery voltage, and open circuit etc. faults should each be identified by a suitable DTC and a unique DTC failure byte describing the error condition. See working example.
 - 93 80 11 2F (Switch Pack) signal short to ground.
 - 93 80 12 2F (Switch Pack) signal short to battery
 - 93 80 13 2F (Switch Pack) signal open circuit

The DTC fault detection shall be confirmed using a counter operation for non-emissions related servers shall be implemented as per the details defined in ISO 14229 D.6 this shall be the criteria for logging/flagging DTCs. The incrementation of the counter shall be determined by the supplier of the ECU based on test/monitoring cycle of the signal and therefore the decrement of the fault occurrence counter, where no faults have been detected it shall be half that of the increment counter. e.g. if the fault detection counter is 30 per test failed (i.e. the test time allocated for this this 20ms) where the server passes the test the decrement of this shall be 15 and the time for dematurity shall be 40ms.

All DTCs that are deployed by the suppliers shall have a debounce strategy based on safety use case derived from FMEA or study conducted. All network and missing message related DTCs shall have strategy based on a minimum two consecutive failed counts of detecting a fault based on safety use case.

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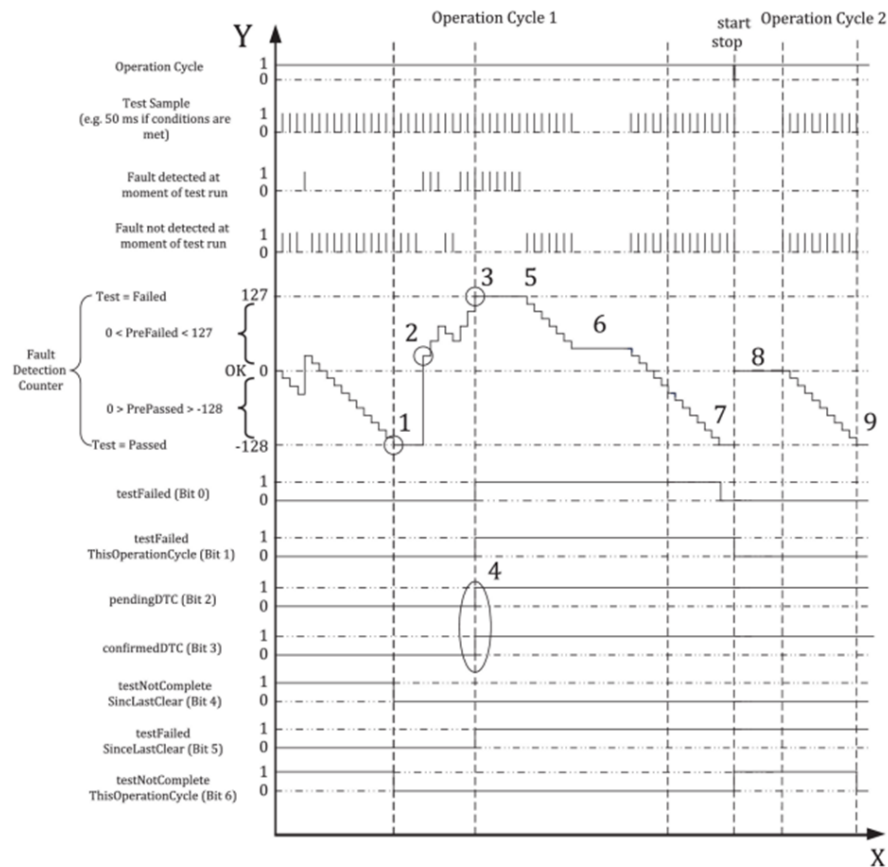


Fig 2

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3.5 On Demand Self-Test (ODST) RID (0303)

The On Demand Self-Test (ODST) the following table contains the details of all the information required for running the routine.

Required for Planning the Routine:		
0x31		
0x0303		
(ODST)		
Application		Flash Boot Loader
Yes		NA
Execute security Level		Execution Time
Level 1/2 if used		Session
		<5s and up to 30s
		10 03/83
To Carry out on Demand Self-Test on the ECU Mandatory for all CAN ECUs		

Data Type	Parameter Name	Cvt	Control Option
	sub-function = [Routine control Type		00 – FF
Control 01	Start	M	NA
Control 02	Stop *see note 1	U/C	NA
Control 03	Results *See note 2	U/C	NA

The routine prior to executing the ODST shall check the vehicle condition is suitable for performing the ODST as stated below: -

Condition for Starting	NRC	Details
Session = Extended	31	Request out of range Routine not available in non-extended session
Condition	22	Condition not correct, unable to execute Routine due to Driver not Present or Ignition state incorrect 0x22.
	92	Routine not able to be executed due to LV battery voltage too high 0x92.
	93	Routine not able to be executed due to LV battery voltage too low 0x93.
	88	Vehicle speed 0x88
Note please see ISO14229 for other NRCs which are applicable for this routine.		

Routine	Request	Response	Details
ODST request	31 01 03 03	71 01 03 03 XX	Where xx = 1 or more 3 Byte DTC codes without the status as per standard DTCs
	31 02 03 03	71 02 03 03	
	31 03 03 03	71 03 03 03 XX...XX	

The ODST is designed to capture failures as they occur, this routine will therefore be looking at all the DTCs supported by the ECU and specifically at the DTC failed status bit, as a fault is detected the DTC test failed status bit changes 0 to 1 this means a problem has been detected and flags a problem instantly, rather than waiting for the DTC to mature, (see table 6). So, if the same DTC was read using service 19 at that moment in time it may not report an issue, this is providing that the DTC has not matured sufficiently, this RID is particularly useful when a part is being fitted/replaced to the vehicle during assembly or in the aftersales, this is to ensure that it is being fitted correctly. This routine shall be supported by every ECU and will be executed via a diagnostic routine that shall detect and identify all of the ECU's (I/O circuit faults (*see note)) that could be affected by incorrect assembly or damage to the ECU or its associated components (actuators, sensors etc.) a misfit which may be caused by Vinfast manufacturing process or after sales service fitment/replacement of the ECU. The routine should use the following guidelines: -

All ECU inputs/outputs should be physically placed in a defined state and verified to be in that state, that employ feedback capability (i.e., all ECU output signals that the ECU can read back) should be energised long enough to detect a fault, and if a failure exists, to identify its type (e.g., open, short to battery/ground, etc.).

Upon completion of the test, the outputs shall return to normal ECU control. An ECU shall complete its on-demand self-test within a maximum time of 30 seconds after receiving the request to perform the test. The routine Control Option Record is not included for this control routine (see reference ISO14229) any additional control option shall be implemented by supplier based on the ECU use case and system behaviour.

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Any failures that are detected and reported by the ODSST shall consist of only three bytes to be able to distinguish these from those reported by service 19 and the failure shall be detected when the ("Test Failed bit 0 is set" see table 6) during the control routine execution which means that the DTC will be reported without reaching maturity.

Note no status of the DTC shall be reported, if the ODSST finds a DTC which has already been detected and its status has been set this routine shall not change the status of any pre-logged DTC when read via service 19.

Table 5A ODSST

Status Of DTC bit	Bit #	Bit State	Description
Test Failed	0	1	DTC failed at the time of the request
Test Failed This Operation Cycle	1	1	DTC failed during the current operation cycle
Pending DTC	2	0	TC failed on the current or previous operation cycle
Confirmed DTC	3	1	DTC is not confirmed at the time of the request
Test Not Completed since Last Clear	4	1	DTC test was completed since the last code clear
Test Failed Since Last Clear	5	0	DTC test failed at least once since last code clear
Test Not Completed This Operation Cycle	6	0	DTC test completed this operation cycle
Warning Indicator Requested	7	0	Server is requesting warning Indicator to be active

Table 6: DTC status

see example below: -

DTC reported using service 19; 94 55 11 1B.

Same DTC reported using service 31 (03 03) 94 55 11.

*Note: - Omitting the evaluation of an ECU I/O circuit shall only be permitted for the following reasons: -

- Testing a circuit would exceed the 30-second on-demand self-test execution time.
- Testing a circuit would require operator interaction during the test.
- Testing a circuit could damage the ECU and/or any of its components.
- Testing the circuit could cause a hazardous ECU/vehicle condition.

For an ECU to qualify for an exemption the supplier will need to show a use case where evidence gathered by FMEA, or risk analysis has shown that running of the routine is likely to lead to an event defined above.

3.6 Snapshot data record

Global/local snapshot, the global snapshot record number captures the global information relating to the vehicle condition and local snapshot captures what is being observed by the ECU locally. The information regarding global/local snapshot shall include all the following as defined in table 7, 8: -

This Global snapshot record number is equal to 0x10 the status on first occurrence; however, ECU can also support DTC record number 11 (Subsequent events) thereby if the failure were to reoccur the two snapshots could be compared to identify if there are any changes in the environment or condition etc. All snapshot DIDs shall be left justified and any remaining data field which remains empty shall be padded with zeros. DTC Snapshot Record Number shall be supported for all numbers valid for the ECU and for 0xFF (meaning report all records). Snapshot records reported by this service shall be reported in single string and in the following order with time stamp followed by Odometer reading then vehicle voltage etc. see working example below.

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Signal name	Cvt	Start byte	Description	DID length in Bits
Time stamp	M	#1	0x F120	48 as defined below
			Date of "Year"	8
			Date of "Month"	8
			Date of "Day"	8
			Date of "Hour"	8
			Date of "minute"	8
			Date of "second"	8
Odometer reading	M	#2	0x F121	24
			odometer	
Supply voltage	M	#3	0x F122	14
			Vehicle Supply voltage	
Ignition status	M	#4	0x F123	8
			Ignition status	
Vehicle Temperature	M/O	#5	DD01* *Mandatory based ECU location	8
			Cabin temperature	
Outside Temperature	M/O	#6	DD02* Mandatory based ECU location	8
			Outside Temperature *Mandatory based ECU location	

Table 7: Global snapshot data record

Working example of Snapshot record: -

Where DD03, DD04 is a local snapshot of the DTC.

19 04 92 8F 11 FF

59 04 92 8F 11 2A 10 08 F1 20 21 10 15 20 30 59 F1 21 00 00 01 00 F1 22 91 F1 23 02 DD 01 19

DD 02 19 DD 03 21 DD 04 78

(Also see ISO14229 table 355)

Local Snapshot record number is equal to = 0x09 to allow for additional local snapshots to be added

Signal name	Cvt	Start byte	Description	Bit length
ECU Temperature	M	#7	0xDD03	8
			ECU temperature	
ECU supply Voltage	M	#8	0xDD04	8
			ECU voltage	

Table 8: Local snapshot data record

Note. The content of table 7,8 shall serve as a bare minimum support for snapshot data but ECU suppliers are encouraged to take a snapshot of whatsoever, they deem necessary to assist them in root-causing the fault/issue. If the ECU has the hardware required to take and report ECU voltage/temperature, then its mandatory for the ECU to report this in the local

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snapshot, if the ECU does not have the necessary hardware, then it should report the global vehicle voltage and temperature.

DD01 Vehicle Cabin Temperature

CAN signal CCU_InteriorTemp
Length = 8 Bits
Unit = Degree C
Value Type = Unsigned
Initial Value = 0
Factor = 0.5
Offset = -40
Min value = -40
Max value = +85

DD02 Outside Temperature

CAN signal CCU_TempEx_AC
Length = 8 Bits
Unit = Degree C
Value Type = Unsigned
Initial Value = 0
Factor = 0.5
Offset = -40
Min value = -40
Max value = +85

Please Note: -In the absence of the temperature/voltage sensors which are missing from hardware design, the supplier must show through FMEA or a feasibility study how they will protect the device.

If in the unlikely event the internal conditions of temperature/voltage or any other conditions were to increase beyond the max or the safe operating range of the device, since operating a device beyond its minimum or maximum operating range is likely to cause stress of the internal components and therefore result in shortening the service life of the ECU.

Time stamp: Dates are always transmitted as Unsigned codes in 6 bytes. The order of the numbers is 'ymmddhhmmss'. The detailed information can be found in the CAN matrix. If the signal is not available from the CAN bus, the maximum value 0x FF FF FF FF FF FF shall be stored.

Odometer reading: The odometer reading will be transmitted as 3 bytes with a resolution of 1 km/bit. The valid range will be from 0 to 16777214. After the maximum valid value (16777214) is reached, the counter will stop and stay at this value. If the bus communication has failed or an invalid odometer value is received, the ECU shall use the last known valid odometer value for the current power cycle, based on CAN spec.

Voltage: Voltages are transmitted as 14 bits with a resolution of 0.0009777 V/bit. This leads to a valid value range of 0 -18V. The maximum value 0x FFFF indicate the signal is not available.

Ignition status: The definition of ignition status signal is in the following:

Signal name: **Ignition status**, Data of length: **3bit**, Value table is as follows:

0x00=Off
0x01=ACC
0x02=On
0x03=Start
0x04~0x06=reserved
0x07=invalid

The DTCs must be stored as **3 Byte Code** and its status for the UDS components including powertrain and chassis ECUs. Therefore, the Supplier of the ECUs must follow the

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requirements of the ISO 15031, ISO 14229 and actual SAE J2012.

Any ECU should have sufficient memory to store as minimum 20 DTCs in a non-volatile memory. The memory must be big enough to store the DTCs inclusive their defined freeze frame data and Local snapshot.

Addition to the above global snapshot information, some other useful information for the error detection is taking a ECU local snapshot.

The requirement of Storage of DTC local snapshot data is specified by the supplier and is specific to each DTC. This data Shall be chosen so that it will be useful in identification of the conditions in which the fault occurred (e.g., operation state and surrounding environment of the ECU) which are present whenever a fault or an intermittent fault (and the fault symptom) occurs.

Please note: - The substituted or default values should not be used for snapshot data as they do not add value or provide any useful information.

3.7 Extended data

Extended data information is a set of data which provides extended status information associated with a DTC; it contains fault occurrence counter, aged counter, aging counter, and priority. The bytes required for extended data record and the byte sequence supported in the "DTC Extended Data Records" of response message are defined in following tables.

This extended record data type which shall capture and report the content of table number 9 as a string proceeding the record number 10, see section 3.13 for a working example of this data record. *Note. The content of table 9, serve as a bare minimum support for extended data record but ECU suppliers are encouraged to maximise on the reporting and gathering of information to be useful when trying to find the root cause of the failure. The content table 9.1 is optional.

Extended data	01-Fault occurrence counter	02- Aged counter	03-Aging counter	04-DTC Severity
Convention value (Cvt)	M	M	M	M
Data length	1 Byte	1 Byte	1 Byte	3 bit
Byte position	Byte 0	Byte 1	Byte 2	Byte3(bit7~bit5)

Table 9: Global extended data record

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3.8 Fault Occurrence counter

The fault occurrence counter indicates how often an error has occurred since the drive cycle has started or last successful DTC clearing was carried out.

This parameter shall be stored in non-volatile memory. Fault occurrence counter will be incremented by 1 when the “test Failed This Operation Cycle” bit is set to “1” from “0”. When this number overflows, it would always equal to 254 and can’t be changed if a new error is occurring. The format of occurrence counter is defined in the following table: 10: (see table 352 in ISO14229)

Data name	Data length	Data type	Resolution	Initial value	Overflow	Fault value/not available value
Frequency of occurrence	1 byte	Unsigned Integer	1	0	254	255

Table 10 the format of occurrence counter

3.9 DTC Ageing management

The deletion of a DTC shall only be possible with diagnostic service if the DTC i.e., not a permanent fault. Also, all related information except aged counter can be deleted autonomously after regular conditions are retrieved, optional combined with a defined amount of ignition cycles or a defined period. System designer should make decision on whether a DTC can be deleted autonomously or not. The number of ignition cycle should be counted by ECUs themselves via clamp signal. Note See section D7 and D11 of ISO 14229 requirement.

3.10 DTC Ageing criteria

A process whereby certain servers evaluate past results of each internal diagnostic to determine if a confirmed DTC can be cleared from long-term memory, e.g. in the event of a calibrated number of failure free cycles. Every DTC which could be aged shall be specified with a number as DTC aging criteria, if during the previous “DTC aging criteria” times consecutive ignition cycles, the test ran and passed in each ignition cycle, a particular DTC will be allowed to be aged. If no other value stated in the DTC list, the value “DTC aging criteria” this shall be equal to 40dec.

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3.11 DTC Severity Definition

- DTCs are entered according to individual Severity. DTCs with a higher Severity overwrite a lower Severity DTC.
- DTC Severity are dictated in several different ways, these Severity are set by the ECU manufacturer to inform you of the severity of the problem. These rank between 0-4 where 4 has the Higher Severity indicating that the issue with higher Severity is more serious and needs to be dealt with immediately. (See ISO 27145-3 ISO14229-1) see D.3 This details the DTC severity and its class definition: -

Every server shall adhere to the convention for storing bit-packed DTC severity information as defined in Table D.11.

The DTC Severity Mask/DTC Severity byte contains DTC severity and DTC class information. The DTC Severity Mask/DTC Severity byte is reported in a 1-Byte value as defined in Table D.11. The upper 3 bits (bit 7 to 5) of the 1-Byte value are used to represent the DTC Severity information. Previously these upper 3 bits have been optional but are mandatory for VinFast BEV Only vehicles. The lower 5 bits (bit 4 to 0) of the 1-Byte value are used to represent the DTC class information.

Table D.11 – DTCSeverityMask/DTCSeverity byte definition

DTC Severity Byte							
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
DTC Severity Information			DTC Class Information				

The DTC severity bit definition defines bit states to report the recommended action to be taken by the system (e.g., vehicle) operator. Table D.12 defines DTC severity status bits.

Table D.12 — DTC severity bit definitions (bit 7 to 5)

Bit	Description	Cvt	Mnemonic
5	maintenanceOnly 0 = no maintenanceOnly severity 1 = maintenanceOnly severity This value indicates that the failure requests maintenance only.	M	MO
6	checkAtNextHalt 0 = do not checkAtNextHalt 1 = checkAtNextHalt This value indicates to the failure that a check of the vehicle is required at next halt.	M	CHKANH
7	checkImmediately 0 = do not checkImmediately 1 = checkImmediately This value indicates to the failure that an immediate check of the vehicle is required.	M	CHKI

The Severity ranking 0-4 is mapped with DTC severity.

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3.12 DTC Class definition

The DTC class definitions apply to OBD systems which comply with the VOBD GTR. Class A, B1, B2 or C are attributes of an emissions-related DTC. These attributes characterise the impact of a malfunction on emissions or on the OBD system's monitoring capability according to the requirements of the VOBD GTR. The DTC class information contained within a diagnostic request is allowed to have more than one bit set to 1 in order to request information for multiple DTC classes. The DTC class information contained within a diagnostic response shall only ever have a single bit set to 1. Table D.13 specifies the GTR DTC Class definition (bits 4 to 0)

Table D.13 — GTR DTC Class definition (bit 4 to 0)

Bit value	Description	Cvt	Mnemonic
0	DTCClass_0 DTCClass 0 is unclassified. This class shall be used if DTCSeverity is included in the response message but no DTC class information is reported, e.g. legacy DTCs as defined in SAE J2012-DA and this document. Bit = 0: DTCClass_0 is disabled for the reported DTC. Bit = 1: DTCClass_0 is enabled for the reported DTC.	M	DTCCCLASS_0
1	DTCClass_1 DTCClass_1 matches the GTR module B Class A definition. A malfunction shall be identified as Class A when the relevant OBD threshold limits (OTLs) are assumed to be exceeded. It is accepted that the emissions may not be above the OTLs when this class of malfunction occurs. Bit = 0: DTCClass_1 is disabled for the reported DTC. Bit = 1: DTCClass_1 is enabled for the reported DTC.	M	DTCCCLASS_1

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Bit value	Description	Cvt	Mnemonic
2	DTCClass_2	M	DTCCCLASS_2
	DTCClass_2 matches the GTR module B Class B1 definition. A malfunction shall be identified as Class B1 where circumstances exist that have the potential to lead to emissions being above the OTLs but for which the exact influence on emission cannot be estimated and thus the actual emissions according to circumstances may be above or below the OTLs. Class B1 malfunctions shall include malfunctions that restrict the ability of the OBD system to carry out monitoring of Class A or B1 malfunctions. Bit = 0: DTCClass_2 is disabled for the reported DTC. Bit = 1: DTCClass_2 is enabled for the reported DTC.		
3	DTCClass_3	M	DTCCCLASS_3
	DTCClass_3 matches the GTR module B Class B2 definition. A malfunction shall be identified as Class B2 when circumstances exist that are assumed to influence emissions but not to a level that exceeds the OTL. Malfunctions that restrict the ability of the OBD system to carry out monitoring of Class B2 malfunctions of shall be classified into Class B1 or B2. Bit = 0: DTCClass_3 is disabled for the reported DTC. Bit = 1: DTCClass_3 is enabled for the reported DTC.		
4	DTCClass_4	M	DTCCCLASS_4
	DTCClass_4 matches the GTR module B Class C definition. A malfunction shall be identified as Class C when circumstances exist that, if monitored, are assumed to influence emissions but to a level that would not exceed the regulated emission limits. Malfunctions that restrict the ability of the OBD system to carry out monitoring of Class C malfunctions shall be classified into Class B1 or B2. Bit = 0: DTCClass_4 is disabled for the reported DTC. Bit = 1: DTCClass_4 is enabled for the reported DTC.		

Note on DTC Severity Byte Usage for BEV Vehicles. Since BEV only vehicles are not yet rated for emissions then the upper 3 bits of the byte must be used so no longer optional in VinFast BEV Only vehicles, along with these bits Bit 0 of the DTC Class table should also be used to Indicate Class = 0 all other Bits 1,2,3,4 should be = 0 since we do not currently report emission related faults.

3.13 DTC Priority Definition

The priority is the combined value of the Severity upper 3 bits and the Class lower 5 bits of the DTC historically the severity bits were optional, so Priority relied on Class of DTC bits 0 to 4 where the bit 1 indicated a higher priority than bit 4, but since BEV has come along Class will not be used apart from bit 0 = 0 class so priority will now be set to align with the severity part of the Byte. So bit 5 when set = 1 and bit 7 when set = 4 and 4= highest priory see table D12.

Example 1 Byte = 0x11 which would mean it's a maintenance only class 0 DTC so low priority

Example 2 Byte = 0x41 which would mean it's a Check immediately class 0 DTC so High priority.

3.14 Aging Counter

This example provides an overview on the operation of a DTC Aging Counter which counts number of driving cycles since the fault was latest failed. (See ISO14229 D.7)

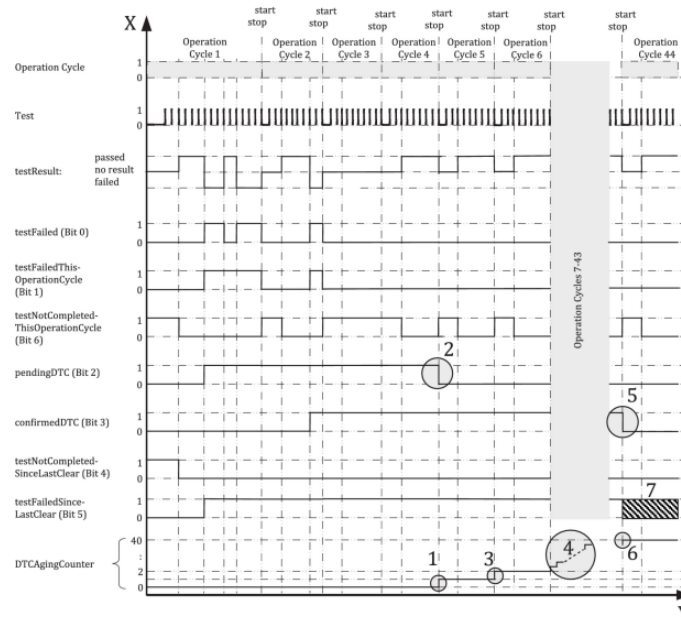


Figure 3 specifies the DTC Aging Counter example.

- DTC Aging Counter is incremented after completing an operation cycle in which test did not fail pending DTC is set to zero after an operation cycle in which the test completed and did not fail. In case an ECU does not support a power down sequence (i.e. is immediately shut off when the ignition is turned off) it will be unable to detect the end of the operation cycle. Therefore, it is also valid to set the pending DTC bit to zero at the beginning of the next operation cycle.
- DTC aging counter counts number of driving cycles since the fault was latest failed excluding the driving cycles in which the test has not reported "test Passed" or "test Failed".
- DTC Aging Counter continues to increment because test is not failing during these operation cycles confirmed DTC is set to zero when aging criteria is fully satisfied (e.g., DTC Aging Counter reaches a specific value).
- DTC Aging Counter reaches a maximum value (e.g. 40) at which time the confirmed DTC bit is cleared it is the responsibility of the supplier to specify whether or not Test Failed Since Last Clear bit is reset by aging-criteria or due to an overflow of the fault memory.

Data name	Data length	Range	Fault/invalid value
Aging counter	8bit	0<aging Counter<DTC Aging Criteria	255

Table 11: aging counter definition

For the aging counter mechanism can refer to (ISO 14229).

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3.15 DTC Aged.

This is a signal which indicates that a DTC has aged sufficiently to be removed from the ECU. If the DTC can't be deleted autonomously, ECU must save the aged counter status 255dec in the error memory.

Data name	Data length	Range	Fault/invalid value
Aged counter	8bit	0...254	255

Table 12: aged counter definition

3.16 Fault Warning Lamp Extinguishing Process.

DTC with lamp indicator shall turn off the fault warning lamp, as part of the DTC aging management.

For non-emission-related DTC with lamp indicator, if during the previous 3 consecutive valid ignition cycles, the fault warning lamp will be allowed to be extinguished, and the "warning Indicator-requested bit" of DTC status byte shall be cleared before it.

3.17 Extended Data Record Handling

Working example of Extended Data being reported: -

0x19 06 92 8F 11 FF

0x59 06 92 8F 11 2E 01 01 02 00 03 01 04 01 10 7F

0x59(Service) 0x06(Sub-function) 0x92 8F 2E (DTC number + status of DTC)

See explanation below: -

0x01 (DTC Extended Data Record Number) 01(Extended data record value for 01) = "Fault Occurrence counter".

0x02 (Extended data record number) 0x00(Extended data record value for Record 0x02) = "Aged Counter this counter will remain reset until it reaches the value determined by the implementor i.e.40dec cycles fault free".

0x03 (Extended data record number) 0x01 (Extended data record value for Record 0x03) = "Aging Counter". This counter value will increment after every complete cycle.

0x04 (Extended data record number) 0x01 (Extended data for 0x04) = "Severity".

0x10 xx (See Fig 2)

Record numbers (see table 9).

Note: Reference [Ref. 1 ISO 14229 table 362] has re-defined the use of this service since the 2006 version of the standard. Previously the standard was not clear about whether DTC Stored Data Record reported the same data as DTC Snapshot Record, but it is now made clear that the data is different. DTC snapshot Data stores data for each DTC from the point of view of the DTC, including the event that caused the storage. i.e., it relates to storage areas that are created dynamically for each DTC fault event. DTC Stored Data stores data related to a fault event from the point of view of the data storage area, including the DTC that relates to the storage. i.e. it relates to a fixed series of storage areas that can be filled with data

The DTC Extended Data records shall be reported according to reference [Ref. 1] and shall not be reported with any service other than service 0x19 Sub-functions 0x06. Note: reporting with service 0x19 sub-functions other than 0x06 is optional. DTC extended data must always be reported with its associated DTC. The extended data has no meaning without its related DTC.

* See table 4

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3.18 DTC Extended Data Additional Explanation

DTC Extended Data counters	Cvt	Size in bytes	Type	Value and data description	Description
0x01	U	1	INT	0-255 count	Record counter 0x01 Cycles since last detected
0x02	U	1	INT	0-255 count	Record counter 0x02 Cycles since last detected in which test completed
0x03	U	1	INT	0-255 count	Record counter 0x03 Cycles since first detected
0x04	U	1	INT	0-255 count	Record counter 0x04 Number of cycles in which fault detected
0x10	U	1	INT	-128- +127	DTC Fault Detection Counter

Table 9.1 DTC Extended data records.

These counters are more intimate than those given in Table 9 where a sequence of operation is fundamental to diagnosing a pattern for DTC fault detection and prediction.

Sequence at the inset (occurrence) of a fault i.e. when fault is first seen by the ECU the counter No 4 starts to count the Cycles in which the fault was present If the fault is present in the following cycle the counter 3 will start to increment along with and following counter 4 If the fault disappears by its own accord then the counter 3 will continue counting and counter 4 will remain static If the DTCs are cleared by the tester all counters are reset.

If then the DTC returns counters 1 & 2 start to count if the fault has completed all action to determine if the fault is Confirmed, then counter 2 will count else counter 1 will count also counter 4 and 3 will count based on ECU drive cycle conditions.

Counter 0x10 basically will read and display the current status of the DTC based on FIG 2. And give a value between +127 = Test Failed 0 = Test OK -128 Test Passed (This also depends on DTC step calibration.)

3.19 DTC Clearing

A DTC can be cleared in two conditions, a diagnostics service request has been sent for a DTC clear or a DTC has aged.

When a DTC is cleared via a clear DTCs diagnostics command, all information connected to that DTC will be reset, i.e., the following parameters will be set to "0":

Fault occurrence counter

Fault pending counter.

Ageing counter

Aged counter

DTC status byte

Global snapshot data

Local snapshot data

Note: - A DTC with "Permanent" (Hard Fault) status cannot be cleared using service 0x14 or autonomously. Since the repair action is to fix the hard fault then run a suitable drive cycle that will verify the repair action has been carried out and then that DTC will be cleared using the internal / autonomous method.

When a DTC is cleared via the DTC ageing mechanism, the same parameters except for the aged counter will be cleared. The aged counter is retained in ECU long term memory to allow the rectifier to determine if that fault was previously deleted due to the ageing mechanism.

3.20 Diagnostic Jobs

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If required, the ECUs must support Services for read and write data and parameter. Logistic data, e.g., HW number or SW number of an ECU, should be supported by all ECUs in the same data format. Production and Service must define their special requirements together with logistic department and diagnostic department.

3.21 Routines for Actuators

All ECUs which are able to actuate internal functionalities with a use of a routine e.g. actuate relays or lamps, etc. should implement such routines so that these functionalities of these ECUs could be checked during Production and Service, the suppliers have to define these and document how to check such functionalities.

3.22 Variant Coding

Relevant ECUs (see component specifications) should be code-able by software and support on this way different versions or functionalities of the ECUs can be actioned. VOD will be a subject of separate document which will capture the required input from ECU owner, suppliers, marketing and other stakeholder about what the features are in the current scope and what features are intended to be proposed for the future and how these are intended to be used in VOD and how the values should be configured for adapting ECU behaviour based on market the vehicle will be sold in and also enable the control of any subscription based feature options.

3.23 Flash / Reprogramming

Relevant ECUs (see component specifications) should support flash functionality. This functionality is needed for a fast and easy change of software during the development process, in Production and in Service.

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3.24 OBDII / EOBD requirements

In case of OBDII/ EOBD relevant tasks, the relevant ECUs has to support this regarding the actual specification ISO 15031 and SAEJ1979 / SAE J1979-DA. Following services must be implemented in the OBD relevant ECUs regarding the requirements of the ISO.

OBD Service	Service Name	Usage
0x01	Request current powertrain diagnostic data	M
0x02	Request powertrain freeze frame data	M
0x03	Request emission-related diagnostic trouble codes	M
0x04	Clear/reset emission-related diagnostic information	M
0x05	Request oxygen sensor monitoring test results	N / A
0x06	Request on-board monitoring test results for specific monitored systems	M
0x07	Request emission-related diagnostic trouble codes detected during current or last completed driving cycle	M
0x08	Request control of on-board system, test or component	M
0x09	Request vehicle information	M
0x0A	Request emission-related diagnostic trouble codes with permanent status	M

Table 13: OBD Services

3.25 Customer Specific Adjustments

The ECUs support additional requirements, based on the specification from Production Team and Service Team. Following requirements exist:

To support the production environment, the control units should support 'production' Mode. All service modes which are necessary to perform the production process must be accessible from this mode.

The supplier must provide a '**production proposal**' description for the control unit in a separate component diagnostic document, which must contain all necessary steps of the production process.

The following tasks are mandatory and need to be explained in detail: -

1. Methods and routines which are required to initialize the control unit.
2. Adjustment or calibration like all around view monitor system.
3. Filling and evacuation
4. Variant Coding
5. MHU FOTA/SOTA implementation Process Procedure
6. OBD related tests
7. Learning process i.e. between key and immobilizer.
8. EOL tests to ensure a full and error-free operation.

The description can for example consist of 'reading of measurement values', 'reading of general data', 'writing of data', 'invoking of InputOutputControls-Routines' and so on.

If some function which is only designed for production purposes, effective locking mechanism must be put in to ensure this function would not be implemented after production.

Any calibration setting that is carried out during production or aftersales using UDS services i.e., 0x31, 0x2E, etc. shall be done behind security unless these are part of the user defined calibration/settings.

3.26 Aftersales Process

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Following the sale of a vehicle if there is module replacement required or it needs to be recalibrated/configured, the following process defines the method by which it can be achieved this is to ensure the ECU behaviour is consistent as intended by Vinfast/ECU supplier.

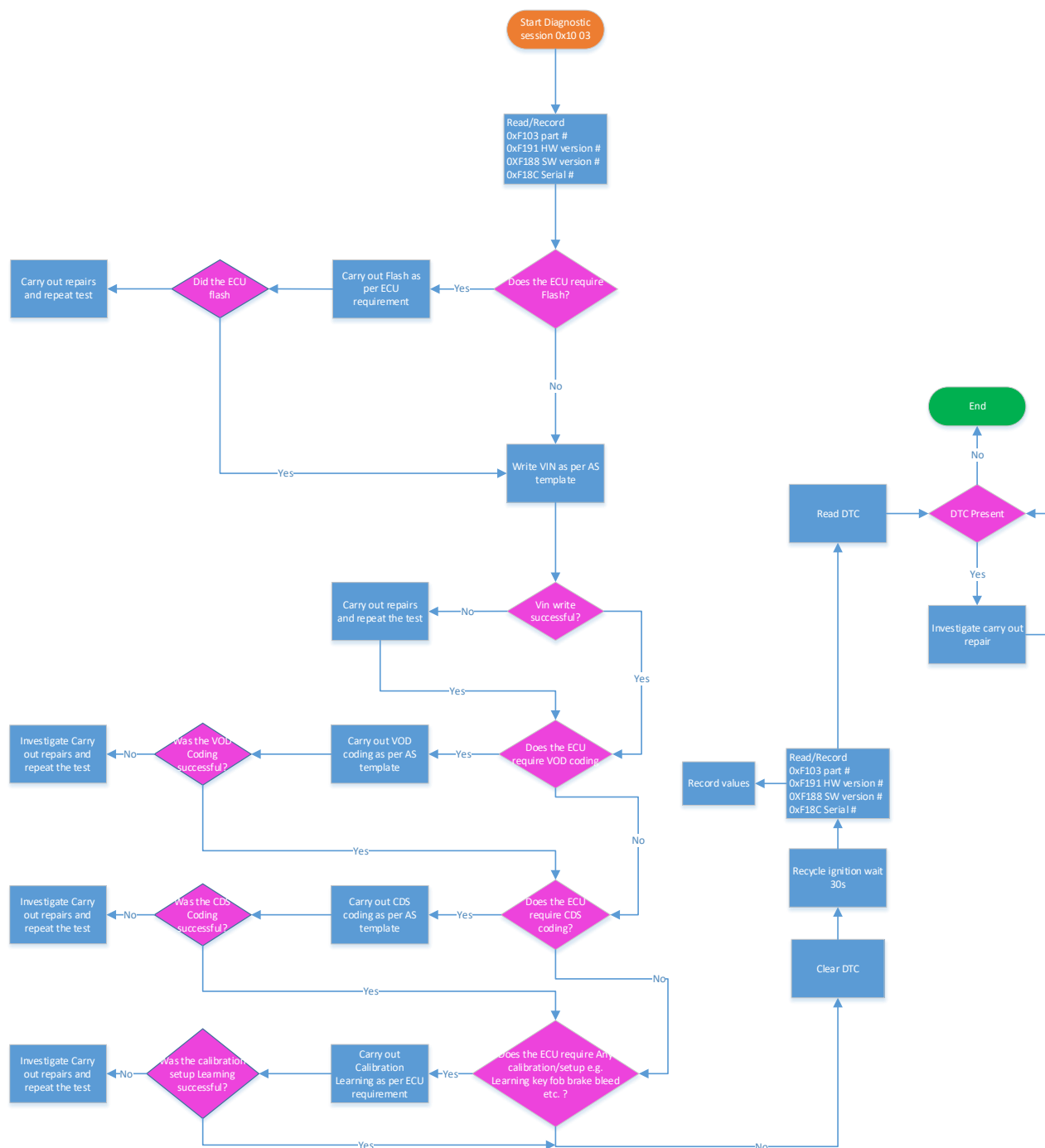


Fig: 4

3.27 End Of Line Test Sequence

The following process shall serve as a EOL sequence that all ECUs which are fitted to the vehicle will need to satisfy. Any calibration or coding sequence which the ECU needs to perform its intended function the supplier will need to detail this in the relevant ODX/EOL/AS documents

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describing how it should be carried out.

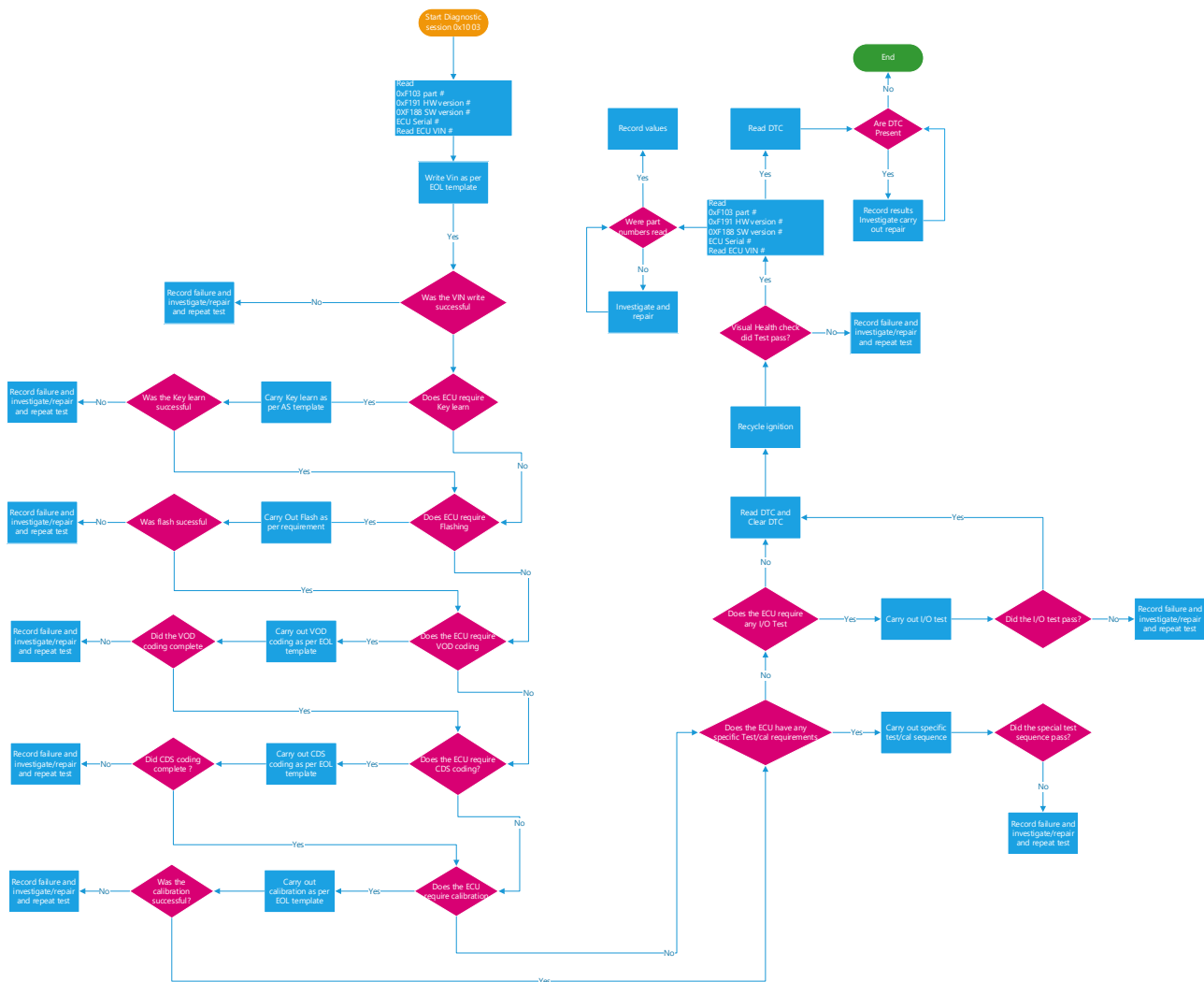


Fig: 5

3.28 LIN Diagnostic

In case of connected LIN ECUs, the Master and Slave ECUs should provide the LIN communication regarding ISO 17987. Dependent of the functionality of the slaves, the supplier of the relevant ECUs must support LIN diagnostic category I, II or III. Detailed information refers to the LIN specification Ref [24].

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3.29 Diagnostic System and Strategy

Overall Diagnostic system and strategy for vehicle faults including capturing, retrieving, and communicating these faults shall support the agreed VinFast customer and connected service models for all markets including any OTA & Remote Diagnostic functions.

Changes in adherence to any diagnostic standards or changes in standards that may lead to a change in compatibility with current or planned diagnostic tool chains must be agreed with Aftersales Engineering prior to the implementation of these standards.

3.30 DTC Selection & Usage (Fault identification and naming)

Think of the service technician who will be using this information to resolve faults. DTC numbers and names need to reflect the serviceability of the system being controlled and the systems it interacts with.

For example

DTC numbering & naming shall reflect industry or VinFast standard definitions where available to allow for:-

- Easy identification of common faults between ECU's.
- Familiarity/consistency across VinFast products.
- Failure-type (Sub-type) shall best reflect the type of fault being diagnosed to allow system faults to be easily identified.
- Any unexpected behaviour caused by any ECU faults shall be diagnosed.
- Communication with other ECU's and any communication system (I.e. CAN-BUS) shall be diagnosed.
- All serviceable components of a system that is sensed or controlled by an ECU shall be diagnosed.
- Embedded components within an ECU or system that are not serviceable do not need to be individually diagnosed but should be covered by single or multiple system level DTCs for the ECU or system.
- If the ECU itself is embedded in a higher-level system/component, then the diagnosis of faults should reflect higher level system DTC's rather than more comprehensive individual sensed or controlled DTC's that would all lead to the same repair action i.e., replacement of the higher-level system/component.

3.31 Service Repair Actions

- It should be assumed that technicians will investigate issues relating with power source/fuses/grounds as part of a general check prior to conducting any specific DTC repair action.
- The repair action shall consider the smallest component that is replaceable as part of any repair action that has been agreed with Aftersales serviceability of the system/component.
- The repair instruction shall only refer to replacing serviceable components, which may be different to production and engineering environments. For example, we shall not instruct the technician to open an ECU and perform electronic repairs/analysis if the internal components are not serviceable and not released as replaceable item by the Aftersales service team. The component owner or Aftersales team should be able to best advice on this topic.
- The repair action shall get to the point where a repair or component/assembly change is required to the vehicle. For example, cycling the ignition is not a repair action.
- Think about what must happen next if this doesn't work? There can be multiple paths/workflows for a given DTC so all options shall be included.

3.32 Input & Output (incl. dynamic DID's) Controls

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All inputs and outputs (physical and logical) at the serviceable system level shall be readable, outputs shall also be controllable where relevant.

3.33 Service Routines

The needs of each ECU and its serviceable system level will determine which routines are required to satisfy its service functions (i.e., replacement, re-programming, configuration, calibration, initialization, learning, self-diagnosis, etc.).

3.34 ECU Traceability and VIN Storage

The ECU requirement of Traceability **shall** be handled through VIN a 17-character unique identification number made of letters and numbers that uniquely identifies vehicle and this **shall** be implemented and stored consistently and correctly within all relevant ECUs such that common EOL and service procedures. This shall be applicable on all ECUs that can support a full or partial diagnostic server functionality and meet one or more of the use cases below: -

- 1 Have a requirement to store VIN to meet requirements of the insurance industry.
- 2 Have a requirement to detect and / or protect against an ECU being moved from one vehicle to another – This will normally be identified during the ECU FMEA process.
- 3 Have a requirement to report the vehicle VIN to meet emissions legislation of various market in which the vehicle is likely to be sold/operated.
- 4 Communicate using I/P, (Ethernet, Broad-R-reach, or some other method of communication between ECUs that is capable of exchanging diagnostic information etc.), and or takes part in the DOIP vehicle discovery process.

VIN Storage Use Cases:

Case 1: Storing VIN to meet the requirements of the insurance industry. An ECU will need to meet this use case if a requirement to permanently store VIN once has been cascaded from Vinfast through to EOL tool connected to the vehicle and thus configuring each applicable ECU to write the VIN using diagnostic commands. The VIN storage requirement is driven by the need to reduce the theft insurance rating of the vehicle by making original VIN traceable by the criminal investigation authorities using diagnostic services and thus allowing stolen and disguised or broken up vehicles to be identified. This is sometimes referred to as the “THATCHAM” requirement. There is no independent diagnostic requirement to store VIN in this way but where the security requirement exists, then there is a diagnostic requirement for the way this VIN storage is implemented.

A VIN learned according to Case 1 shall be readable with DID 0xF190 using the format defined in the Reference document, (Basic DID requirement). Prior to storing a VIN, F190 shall read all 0x00 (ASCII Null character). Note: DID F190 reports 24 characters using ASCII format and the VIN only uses the first 17 of these characters. In accordance with the requirements for data type ASCII. A VIN learned according to Case1 shall not be used for any other purpose by the ECU. The ECU shall remain fully functional regardless of whatever value of VIN is stored according to Case1, whether it matches the vehicle VIN or not, or even in the case if a VIN has not been stored at all. No DTCs shall be recorded/logged for a VIN mismatch since there will be no customer observable symptom associated with misaligned VIN.

Case 2: Storing VIN as a method of detecting if an ECU has been moved between vehicles. An ECU will need to meet this use case where an FMEA or risk analysis has shown that moving an ECU that had been set up to operate on a specific vehicle to another vehicle might result in undesirable ECU behaviour. For example, if a steering angle sensor had been calibrated to the straight-ahead position in one vehicle and if then the sensor ECU was moved to another vehicle

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the calibration would become invalid and therefore, ECUs using the sensor data may then function incorrectly. A method to prevent this is to store the VIN in a (DID 0x0803) at the point an ECU is set up and then confirm the vehicle VIN is unchanged each time the ECU operation cycle is started this will only apply. Where there is a requirement to identify if an ECU has been calibrated or locked to a specific vehicle and then been moved to another vehicle then the ECU calibration or lock process shall be designed to store all or part of the vehicle VIN in the non-volatile memory as part of that process. (Currently the whole VIN is not transmitted by the BCM hence only the last characters of the VIN will be compared to ensure that the ECU is still connected to the same vehicle). This stored VIN shall not be readable with DID F190. Each calibration / lock / build stage process that can be invoked separately shall store its own copy of the VIN and shall not be shared with any other ECU function that uses VIN including any storage under Case 1.

It maybe, the case where it's preferable to show the VIN that the ECU vehicle is locked to or where the calibration was last learned in which case the calibration VIN shall be stored in a different DID, under normal circumstances both DIDs shall mirror each other, therefore any ECU that needs to store linked VIN to calibration will need to subscribe to VIN networked ECU that have stored a VIN according to Case2 and which have a functional requirement to detect the ECU being moved between vehicles shall check this stored value against the transmitted VIN value as soon as possible after the start of each vehicle operation cycle once new VIN data has been correctly received. If a mismatch is detected the ECU shall take action according to its functional specification. If that action includes setting a warning visible to the vehicle driver, then a DTC shall be set that indicates the system is not calibrated or that an incorrect ECU is fitted. If the action is that recalibration is carried out automatically in the background, then if customer symptoms or warnings are present during this automatic process a suitable DTC shall be set but otherwise no DTC shall be set. The DTC shall not state that the VIN is incorrect only to indicate that the calibration is required of the affected ECU otherwise this will prompt an investigation which is not needed.

ECUs that have stored a VIN according to Case 4 shall determine a policy of whether the ECU is allowed to be updated / recalibrated to align with the new vehicle or whether once locked to a vehicle the ECU is never allowed to operate in another vehicle. In either case suitable control routines shall be implemented to store the VIN either once when the ECU is first fitted / calibrated or every time the calibration is re-run as part of the calibration process or as part of a control routine that exits a plant build mode or other similar special operation mode. ECUs shall not be implemented to store the VIN automatically (e.g. on first ECU power up) except where an associated re-calibration is also triggered automatically.

Note: where the ECU is locked permanently to a vehicle, consideration should be given to the cost of scrap ECUs when recovering from misguides where vehicles are initially loaded electronically with a VIN that does not match the physical VIN plates.

Case 3: - Storing VIN so it can be reported to OBD scan tools as required to meet emission legislation.

An ECU will need to meet this use case where the ECU has a separate requirement to meet the SAE J1979 specification. Only a small sub-set of ECUs will have this requirement. ECUs with this requirement need to store the vehicle VIN that is (periodically transmitted over the vehicle network) so that this can be reported immediately to a scan tool without the need to wait for this periodic transmission.

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This requirement is applicable only to emission relevant ECUs that are required to support legislated OBD scan tool communication according to SAE J1979 and are required to support OBD mode/service 0x09 info-type 0x02 (VIN). The Case is not applicable otherwise, ECUs that meet the criteria above shall store all 17 characters of the vehicle VIN as part of the vehicle configuration. Typically, ECU which fall in this category this will require storage of any changed data (like case 2) at power down and first use of the updated data in the next operation cycle. This check/update shall occur automatically on power down/shut down of the ECU. Apart from reporting this VIN using OBD mode/service 0x09 info-type 0x02 the same VIN shall also be readable using UDS (DID 0xF800) as defined in reference (12)

Case 4: - Storing VIN so it can be reported to a diagnostic tool during the DOIP vehicle discovery process. An ECU will need to meet this use case where the ECU supports diagnostic communication over internet protocol (DOIP) and has been designated as the DOIP edge node or a DOIP entity that takes part in the DOIP vehicle discovery process according to Reference Ref. (ISO 13400). ECUs with this requirement need to store the vehicle VIN that is periodically transmitted over the vehicle network (TBD) so that it can be reported immediately to a tester as part of vehicle discovery without the need to wait for this periodic transmission.

An ECU that has been designated as a DOIP edge node or a DOIP entity such that it takes part in the DOIP vehicle discovery process shall store all 17 characters of the vehicle VIN. Typically, this will require storage of any changed data at power down and first use of the updated data in the next operation cycle. This check / update shall occur automatically on power down/shut down of the ECU. A VIN stored to meet this requirement shall use a storage area separate from any VIN storage to meet Case 1 or a ECUs that support a bootloader, the VIN storage area shall be accessible by the ECU bootloader, as specified by the implementer is permitted to share the VIN stored to meet this Case with storage requirements implemented to meet use case 4 so long as no conflict between the use cases has been identified. Where the VIN2 storage is shared in this way the stored VIN will be readable using (DID 0xF801) as required by SAEJ1979. In cases where the VIN storage cannot be shared and optionally in all other cases the VIN4 shall be readable using (DID 0xF802).

Note: Case 4 This requirement is only expected apply to ECUs that are connected to a network that can support TCP or UDP Ethernet protocols.

3.35 Programming in Service

For any ECU that is re-programmable in service either for ECU replacement or for software updates, all data files necessary for programming must be released in EBOM (i.e. flash driver, software and calibration files). Even if these files are not used by ENG or MFG. Part number and version control of these files must follow VinFast spec books and guidance documents, particularly around changes from saleable builds and beyond. Identification and storage of these files must follow agreed methods for Aftersales service and OTA.

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3.36 ECU Diagnostic Information Templates

ECU diagnostic & programming information shall be provided in a format approved by Vinfast. Must be in English language only. Must include the following DTC information layout:

Property	Type	Description	Usage
DTC Number (hex)	3 byte	Per ISO15031	M
DTC Number	String	Per ISO15031 conversion for 3-byte format (eg 0xC14087 = U0140-87)	M
Failure Type Byte	1 byte	Per ISO15031/SAE-J2012DA	M
DTC Name	String	Description of the DTC + Failure Type Byte Description	M
Severity	Upper 3 bits (7 - 5) Enumeration	DTC severity information – These Bits are Mandatory for VinFast BEV only vehicles	M
Class	Lower 5 bits (4 – 0) Enumeration	0 - DTC Class 0 (only used if Severity used) Expect =1 1 - DTC Class_1 matches the GTR module B Class A definition Expect = 0 2 - DTC Class_2 matches the GTR module B Class B1 definition. Expect = 0 3 - DTC Class_3 matches the GTR module B Class B2 definition Expect = 0 4 - DTC Class_4 matches the GTR module B Class C definition Expect = 0 Reference DTC severity- ISO 27145-3	M
Lamp Flag	Boolean	Indicates if a lamp indicator is requested (or driver information warning message is displayed)	M
Repair action	String	A textual description/workflow of how to solve the problem, related to the DTC.	M
Monitor enable conditions	String	A textual description of the conditions, required to execute the DTCs monitor correctly.	M
Monitor type	Enumeration	Defines whether the DTCs monitor is executed either continuously or by an event.	M
Monitor rate	String	A textual description on how often a DTC is checked (continuously), or when the DTC is checked (based on an event).	M
Test Failed criteria	String	A textual description of the conditions for the DTC monitor to produce test result "Test failed ".	M
Mature Time	String	How long fault is present before DTC is triggered	M
Test Pass criteria	String	A textual description of the conditions for the DTC monitor to produce the test result "Test passed ".	M
Service relevant	Enumeration	De-fines whether a DTC is relevant for service (Yes/No)	M
Demature Time	String	When all test pass criteria are met again, how long does it take for a Current DTC to transition to History status	M
Limp-home action	String	A textual description (required for service) of the remedial action applied to the performance of the ECU, component, or vehicle.	U
DTC Aging criteria	String	A textual description of the conditions or cycles required to appraise the DTC as aged	M
Local snapshot record	String	The information should be stored when the DTC occurrence	U
Vehicle Safety Level	String	0 - Normal 1 - Slight Fault 2 - Minor Fault 3 - General Fault	U

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		4 - Serious Fault 5 - Fatal Fault		Table 15: Layout of DTCs
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All additional amounts for the DTC implementation, will be defined by Diagnostic department, based on ISO/SAE conformity.

4.0 Diagnostic Communication

4.1 Communication Speed

The Bus speed of Diagnostic-CAN is 500kbps.

4.2 Can Network Layer Parameter

For CAN Network Parameters please see CAN specification requirement (VFDSXVEEP0047).

4.3 Diagnostic Layer Parameters

Diagnostic layer Parameter	Min	Max	Timeout	Boot	Unit
P2server	0	50	n/a	25	ms
P2client	n/a	n/a	150	150	ms
P2*server	0	2000	n/a	2000	ms
P2*client	n/a	n/a	5000	5000	ms
S3 server	n/a	n/a	5000	5000	ms
S3client	0	3000	4000	4000	ms
P4 CAN Server	0	5000	5000	5000	ms
Network layer Parameter	Timeout Value		Performance requirement		
N_As/N_Ar	70ms		--		
N_Bs	150ms		--		
N_Br	--		$(N_Br + N_Ar) < 70ms$		
N-Cs	--		$(N_Cs + N_As) < 70ms$		
N_Cr	150ms		--		
ECU Flow Control Parameters	Application Mode		Boot Mode		
BS	8		0		
ST min	20ms		0ms direct and up to 5ms max on the Vehicle going through XGW depending on the min process time of the XGW.		

Table 16: Requirement of Diagnostic and Network Layer Parameter (note #)

Note: timing requirement for the above table taken from (ISO-15765-4 ISO-14229-2 ISO-15765-2) and shall always be referred for implementation where required.

S3 Server time: -

Keep a diagnostic session other than default session active whilst not receiving any diagnostic messages for a maximum of 5000ms time out. S3Server timer handling is based on the network layer service primitives, which means that the S3Server timer is also restarted upon the reception of a diagnostic request message even if that message is not supported by the server.

S3 Client: -

Time between functionally addressed Tester Present (0x3E) request messages transmitted by the client to keep a diagnostic session other than the default Session active in multiple servers (functional communication) or maximum time between physically transmitted request messages to a single server Physical communication). Min 0ms Max 3000ms Time out 4000ms.

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S3 Timer: -

Where there is communication between ECU and tester as long as the time between communication does not exceed the P2*_server_max time of **2000ms** it should maintain session and reset the S3_timer after each communication that is able to reach the network layer, this should be carried out even if the service is not supported by the ECU. Please refer to ISO14229-2 for clarification.

Any time a server that is in the process of handling any diagnostic service, it stops its S3 timer. When the diagnostic service is completely processed then the server restarts its S3 timer.

P2 Server

Performance requirement for the server to start with the response message after the transmission of a negative response message with response code 0x78 (enhanced response timing). The ECU must respond with either a positive or negative response within a maximum **5000ms** even if the message was sent with suppress positive response bit set.

Time between tester request message and the receipt of an un-segmented response message or first frame of a segmented response message

The ECU shall start sending its response message within P2CAN after the Request message has been correctly received. P2CAN_Server 50ms.

4.4 Diagnostic services response times

An ECU shall always execute and complete a request before transmitting a response unless specified otherwise. If the time required to complete a request is longer than the specified value P2CAN Server, then the ECU may respond with a busy repeat request (Negative Response Code NRC 0x21) or NRC 0x78 “request Correctly Received Response Pending” as a method of delaying its final response. The timing parameter P2CAN_Server is specified in table 16. Note that the maximum value of P2CAN_Server is fixed and it's the same in the default Session and extended Diagnostic Session.

If a diagnostic Session Control (0x10) or ECU Reset (0x11) request requires a reset of the ECU, the positive response shall be made before carrying out the requested action. Until the requested action is complete the ECU shall either ignore any received diagnostic requests or respond with a negative response code of Busy Repeat Request (0x21). Note: see ISO14229-2 2013 which specifies the maximum time during which the ECU is permitted to be unresponsive to -request messages following a reset of the ECU. Generally, a CAN ECU should be ready within 500ms after it has given a positive response to reset request.

If a diagnostic Session Control service 10) or ECU Reset (0x11) request does not require a reset of the ECU. The ECU shall complete the functionality simultaneously with or immediately following the positive response so that the requested functionality is guaranteed to be complete prior to the arrival of the next possible diagnostic request by the application.)

If a request is sent with the Suppress Positive Response Message (SPRM) Indication Bit set to true, the ECU shall execute and complete the request within P4CAN server time (5000ms). Note that when negative response code 0x78 is utilised for a given request, the ECU shall always send its final response to that request (i.e., positive response or negative response message with negative response code other than 0x78), which is independent from the value of the PRM Indication Bit.

4.5 Diagnostic Layer Error Handling

Error handling for the application layer and session management to be fulfilled by the client and server during physical and functional communication shall be in accordance with the ISO 15765- 3.

The client error handling defined shall be performed for a maximum of two times, which

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means that the worst case of service request transmissions is three.

Additionally, if there is a server timeout, the server must return to default session immediately. When an ECU is sent a request that it is processing it should not reply back with response back unless it is ready to process the next request, exception to this requirement is when the request is for an ECU reset in which case the ECU is required to send the response and then carry out the reset. Whilst recovering from reset ECU shall not respond to any further requests unless it is ready to process the request and it should not buffer any requests. This is to prevent the ECU from buffering requests as multiple requests from the client may result in the ECU being overwhelmed and may cause it to lock up. The maximum period for a CAN ECU to be non-responsive following a reset is 500ms.

4.6 Functional Addressing

Functional addressing Servers All the requests with length less than eight can be physically or functionally addressed.

- All the requests with length greater than seven must be physically addressed.
- Where Servers do not support the request replace the negative responses containing NRCs 0x11, 0x12, or 0x31 with "No Response" for all functionally addressed requests since these NRCs add no value.
- Where the Server response is greater than 8bytes then the response shall be given in a segmented response.

4.7 Data length code (DLC)

The CAN DLC contained in every diagnostic CAN frame shall always be set to eight (8). The unused data bytes of a CAN frame shall be padded with zero (0). Any diagnostic CAN frame with a DLC less than eight (8) shall be ignored.

Any functional request CAN frame with dynamic DLC can be less than defined DLC of fixed length. Where a packet is missing will lead to ECU declaring missing message error.

4.8 Addressing method

For the current Project follow addressing method is required:

- Normal Addressing

For this method the requirements of the ISO 15765 are binding. The using of the method is described in the ISO 14229 for UDS ECUs. Deviations must be coordinated with the Project Architecture and Diagnostic Team.

4.9 ECU Address

Each ECU in CAN network has one specific address. This is necessary to send the service request to the correct ECU. For special services, a functional address (0x6FF) will be used. This broadcast address is needed for requests to all ECU's (e.g., UDS Service0x19 use a functional address, to read DTCs from all ECU's in the network with sending the request only one time). The addresses of all ECU's in the network must be defined by the network manager.

In case of different components and communication protocols on the CAN Bus systems the addressing method and calculation of Response IDs are different. For further information, refer to the document below:

- ECU_Identifier_Specification

For OBD CAN Identifiers the requirements of the ISO 15765 (part 4) are

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binding. Deviations shall agree by the Project department.

5 System Description

5.1 General System Description

The BUS Architecture include different BUS systems. For further information, refer the document:

System_schematics for the vehicle.

5.2 CAN Network

For the CAN network the requirements of the CAN Network Specification is binding:

- CAN_Specification [ref 6]

5.3 LIN Network

For the LIN network the requirements of the LIN Specification are binding for further information refer the document:

- LIN Specification ISO 9141-2 for guidance

5.4 Specific Requirements

All ECUs in the Project must store internally information about Hardware, Supplier, Software, ECU serial number etc. This are common Data Identifier. For further information, refer the document:

- Basic Services DID PID Definition

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6 General Diagnostic Specification

6.1 Diagnostic Protocol

The communication between car and tester must comply with the requirements of Unified Diagnostic Services UDS ISO 14229-1-2020.

A special part of the ON-Board diagnosis is the fault detection of emissions-related parts regarding OBDII/EOBD. Here are the special requirements of the ISO 15031 (part 1 ... part 7) are binding.

6.2 Diagnostic Data

The defined data exchange format for all ECUs is an ODX package. This data package must include all necessary diagnostic information of the ECUs. The Supplier must deliver a fully tested file regarding ODX standard ISO 22901 as Version 2.2. The Supplier makes sure, that the file is running and can be used on a valid MCD Server regarding ISO 22900.

For special tasks like Flash and Coding support, the supplier delivers a fully functional ODX data file.

6.3 Diagnostic Sessions

For diagnostic task, the ECU must support different sessions. The is therefore required to inform the tester when requested about the session the ECU is in via DID F186. This chapter introduces the different diagnostic sessions, which should be available for the relevant ECUs. Requirement about the common Service and Subservice are described in the ISO14229 and CES Basic Services DID PID Definition.

6.4 Default Session

The Default Session shall be enabled in the ECU, if no other session is requested after power up.

6.5 Extended Session

The Extended Session shall be enabled in the ECU, if this diagnostic task is requested by the tester. This session shall be maintained for the duration of the P2*server_max time out of 2000ms unless a subsequent request is sent which will reset the timer before P2*Server_max timeout occurs. Please note any diagnostic request can be used to reset the S3 timer to maintain the current non default session but server must respond with a valid or invalid response within the P2*Server time for the tester to continue with diagnostic.

6.6 Coding Session

The Coding Session shall be enabled if the Coding process requested by the tester. Please refer to the coding specification for the specific details. This session shall be maintained for the duration of the P2*Server_max time out of 2000ms unless a subsequent request is sent which will reset the timer before P2*Server_max timeout occurs. Please see the respective P2*Server time relating to Bootloader.

6.7 Programming Session

The p2shall be enabled, if the Flash/Programming sequence is requested by the tester, please refer to flash specification for specific details.

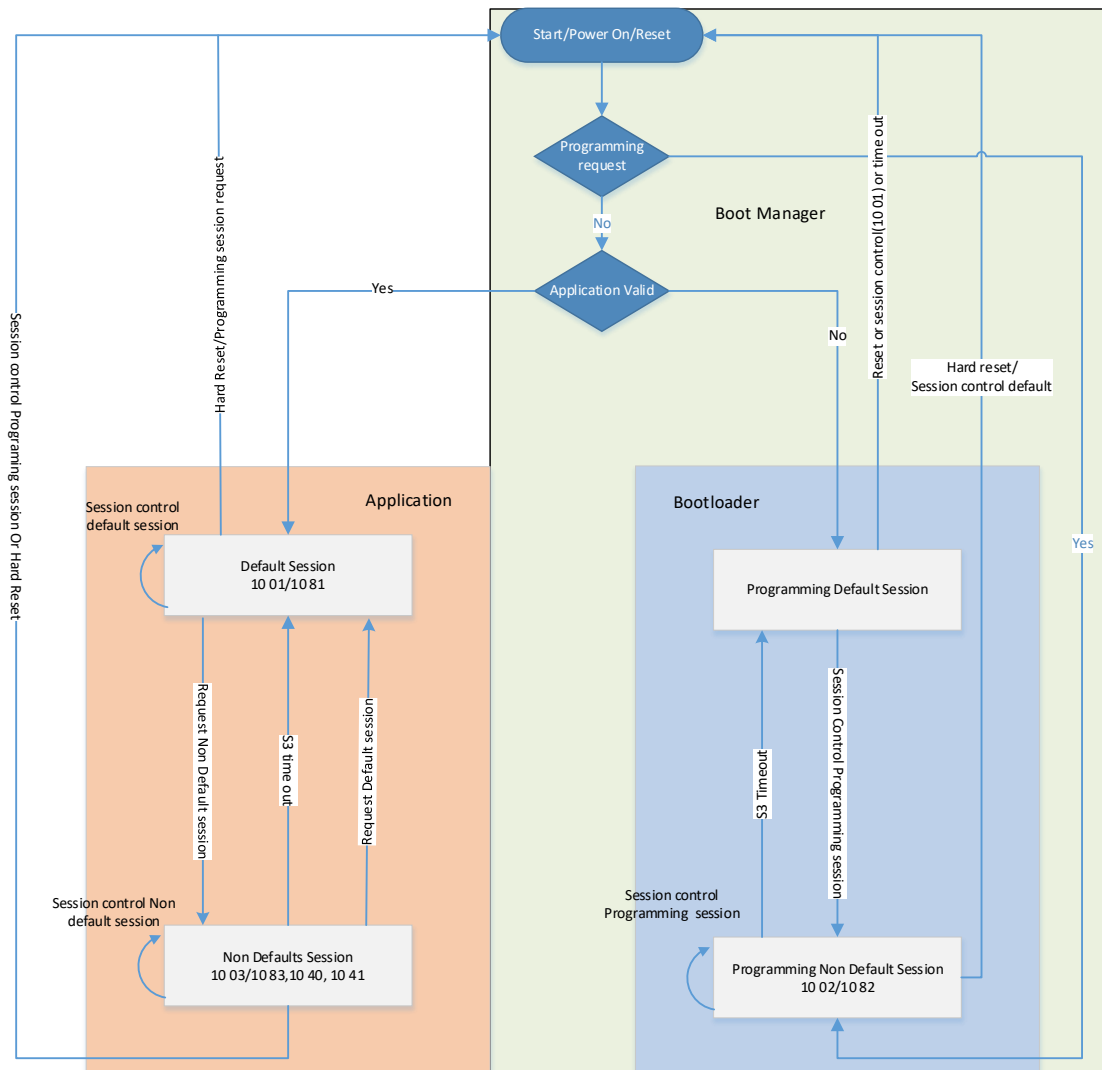


Figure 6: Session Control

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6.8 Security Level for ECU Access

To protect against unauthorized access to the vehicle electronic system, the service Security Access (0x27) method shall be used.

For each of the Security Levels defined in table 12 and table 13 below, the Security Access (0x27) Keys will be unique

Security Level	Description	Key must defined by OEM	Usage
0	No Level needed or OBD relevant amounts	no Key needed	M
1	After Sales (Service)	RND and OEM Key	M
2	Manufacturing	RND and OEM Key	M
3	ECU-Coding	RND and OEM Key	O
4	ECU-Programming (Flash)	RND and OEM Key	O

Table 17: Security Level

Security Level	Description	Request Seed	SendKey	Session	usage
0	No Level needed or OBD relevant amounts			Default	M
1	After Sales (Service)	0x01	0x02	Extended	M
2	Manufacturing	0x03	0x04	Extended	M
3	ECU-Coding	0x05	0x06	Coding	O
4	ECU-Programming (Flash)	0x07	0x08	Programming	O

Table 18: Seed and Key Subservices

For full Security Access (0x27) specific requirements, please refer to VF_Security_Access_Algorithm specification.

6.9 Diagnostic Services

The diagnostic service is the summary of a request message and a response message. Any diagnostic service has an explicit service identifier.

- Type request
- Type response
- Type positive response.
- Type negative response.

Each service request by a Tester must be answered by the ECU with a valid response. It is not allowed, that the ECU does not answer to the Tester request.

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6.10 Diagnostic Service Example Simple Communication

Simple Communication in UDS without Sub- Function Byte

Example

- 1 Positive Response
- 2 Negative Response

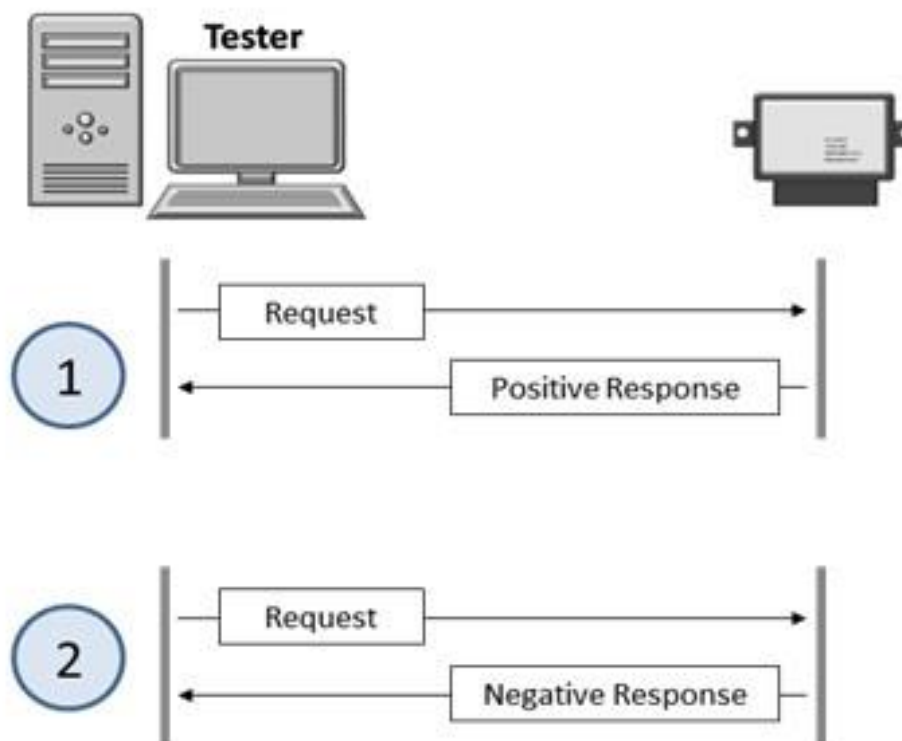


Figure 7: Simple Communication

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6.11 Diagnostic Services UDS

This chapter describes the Diagnostic services for the UDS protocol.

All mandatory services in this chapter shall be supported by the ECU supplier. The ECU Shall support suppress positive response requests in all services where this is deemed helpful; this is particularly useful where the use of functional addressing is required to carry out the same task across the whole vehicle with just a single command i.e. to change the session of the whole vehicle or to clear DTCs etc. An overview about the requested services is in the table 19 below:

SID hex	Service Name	Usage
0x10	DiagnosticSessionControl	M
0x11	ECUReset	M
0x27	SecurityAccess	M
0x28	CommunicationControl	M
0x3E	TesterPresent	M
0x83	AccessTimingParameter	U
0x84	SecuredDataTransmission	U
0x85	ControlDTCSetting	M
0x86	ResponseOnEvent	U
0x87	LinkControl	U
0x22	ReadDataByIdentifier	M
0x23	ReadMemoryByAddress (will not be allowed due to security concerns)	NA
0x24	ReadScalingDataByIdentifier	U
0x2A	ReadDataByPeriodicIdentifier	U
0x2C	DynamicallyDefineDataIdentifier	U
0x2E	WriteDataByIdentifier	M
0x3D	WriteMemoryByAddress (will not be allowed due to security concerns)	NA
0x14	ClearDiagnosticInformation	M
0x19	ReadDTCInformation	M
0x2F	InputOutputControlByIdentifier	U
0x31	RoutineControl	M
0x34	RequestDownload	U
0x35	RequestUpload (will not be allowed due to security concerns)	NA
0x36	TransferData	U
0x37	RequestTransferExit	U
0x29	Authentication	M
0x38	File Transfer where ECU supports file transfer	M

Table 19: Table of Services ISO 14229 UDS

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6.12 0x10 DiagnosticSessionControl

The client requests to control a diagnostic session with a server(s).

Diagnostic Session Control Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	Diagnostic Session Control Request Service Id	M	0x10
Sub Function = []			
Data Byte No	Parameter Name	Usage	Data Value
2	sub-function = [diagnosticSessionType]	M	0x00 - FF
	defaultSession	M	0x01 /81
	programmingSession	M(CAN node)	0x02 /82
	extendedDiagnosticSession	M	0x03 /83
	codingSession	M	0x41/C1
	Bench session	U	0x40/C0
	Roller Session	U	0x50/D0

Table 20 Diagnostic Session Control

6.13 0x11 ECU Reset

The client forces the server(s) to perform a reset, the server will send a response back and then carry out the reset as requested, whilst undergoing the reset the server shall be not responsive for a duration no longer than 500ms.

ECUReset Resquest Service Id0x11			
Data Byte No	Parameter Name	Usage	Data Value
1	ECUReset Request Service Id	M	0x11
subFunction = []			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [resetType]	M	0x00 - FF
	hardReset	M (programming)	0x01/81
	keyOffOnReset	C	0x02/82
03	softReset	M	0x03 /83

Table 21 ECU reset

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Definition of Types of Resets

The ECU-Reset service is used by the client to request a server (ECU) to reset. Below is the list of resets that are used in service 11 and the expected ECU behaviour following such a request

01 = Hard reset, this simulates the power off from the ECU performing this which will erase anything in the volatile memory and the reapplying power the ECU will perform start up sequence. This is a mandatory service and must be supported by all ECUs

02 = Key off-on reset This value identifies a condition similar to that of the driver turning the ignition key off and back on. This reset condition should simulate a key-off-on sequence (i.e. interrupting the switched power supply). The performed action is implementation specific and not defined by ISO 14229. Typically, the values of non-volatile-memory locations are preserved; volatile memory will be initialized.

03 = soft reset. This value identifies a “soft reset” condition, which causes the server to immediately restart the application program if applicable. The performed action is implementation specific and not defined by ISO 14229. A typical action is to restart the application without re-initializing of previously learned configuration data, adaptive factors and other long-term adjustments.

04 = Enable rapid power shutdown This value requests the server to enable and perform a “rapid power shut down” function. The server shall execute the function immediately after “key/ignition” is switched off. While the server executes the power down function, it shall transition either directly or after a defined stand-by time to sleep mode. If the client requires a response message and the server is already prepared to execute the “rapid power shut down” function, the server shall send the positive response message prior to the start of the “rapid power shut down” function. The next occurrence of a “key on” or “ignition on” signal terminates the “rapid power shut down” function. The client shall not send any request messages other than the ECU-Reset with the sub-function disable Rapid Power Shut-Down in order to not disturb the rapid power shut down function.

NOTE this sub-function is only applicable to a server supporting a stand-by mode.

05 = Disable rapid power shutdown. This value requests the server to disable the previously enabled “rapid power shut down” function.

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6.14 0x27 SecurityAccess

The client requests to unlock a secured server(s).

Security Access Request Service Id0x27			
Data Byte No	Parameter Name	Usage	Data Value
1	SecurityAccess Request Service Id	M	0x27
subFunction = [securityAccessType = requestSeed]			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [securityAccessType = requestSeed]	M	0x00 – FF
	requestSeed (afterSale)	M	0x01
	requestSeed (EOL)	M	0x03
	requestSeed (Coding)	M	0x05
	requestSeed (flashing)	M	0x07
subFunction = [securityAccessType = sendKey]			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [securityAccessType = sendKey]	M	0x00 – FF
	sendKey	M	0x02
	sendKey	M	0x04
	sendKey	M	0x06
	sendKey	M	0x08
3	securityKey[] = [key#1 (high byte)	M	0x00 - FF
:	:	:	:
#n	key#m (low byte)]	U	0x00 - FF

Table 22 Security Access

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6.15 0x28 CommunicationControl

The client requests the server to control its communication.

CommunicationControl Request Service Id0x28			
Data Byte No	Parameter Name	Usage	Data Value
1	CommunicationControl Request Service Id	M	0x28
subFunction = [controlType]			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [controlType]	M	0x00 - FF
	enableRxAndTx	M	0x00 /80
	enableRxAndDisableTx	M	0x01 /81
	disableRxAndEnableTx	U	0x02 /82
	disableRxAndTx	M	0x03 /83
communicationType			
Data Byte No	Parameter Name	Usage	Data Value
3	communicationType	M	0x00 - FF
	normalCommunicationMessages	M	0x01
	networkManagementCommunicationMessages	M	0x02 *
	NormalandNetworkManagementCommunicationMessages	M	0x03*
* 1, If an ECU implements network management function,0x02 shall support this communication type for communicationControl service. 2, For flashing function, 0x03 should be supported by all the ECUs.			

Table 23 Communication control

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6.16 0x3E TesterPresent

The client indicates to the server(s) that it is still present. All servers shall support this service when it targeted at the ECU ID as well as the ECU functional ID with both positive response and suppress positive response.

TesterPresent Request Service Id0x3E			
Data Byte No	Parameter Name	Usage	Data Value
1	TesterPresent Request Service Id	M	0x3E
subFunction = [zeroSubFunction]			
Data Byte No	Parameter Name	Usage	Data Value
2	sub-function = [zeroSubFunction]	M	0x00 - 7F
	zeroSubFunction	M	0x00 /80

Table 24: Tester Present

6.17 0x83 Access Timing Parameter

The client uses this service to read/modify the timing parameters for an active communication.

AccessTimingParameter Request Service Id0x83			
Data Byte No	Parameter Name	Usage	Data Value
1	AccessTimingParameter Request Service Id	U	0x83
subFunction = [timingParameterAccessType]			
Data Byte No	Parameter Name	Usage	Data Value
2	sub-function = [timingParameterAccessType]	M	0x00 - 7F
	readExtendedTimingParameterSet		0x01
	setTimingParametersToDefaultValues		0x02
	readCurrentlyActiveParameters		0x03
	setTimingParametersToGivenValues		0x04
timingParameterRequestRecord			
Data Byte No	Parameter Name	Usage	Data Value
3	TimingParameterRequestRecord[Byte#1	C	0x00-FF
	:	:	:
#n	Byte#m]	C	0x00-FF

Table 24 Access Timing

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6.18 0x85 Control DTC Setting

The client controls the setting of DTCs in the server.

ControlDTCSetting Request Service Id0x85			
Data Byte No	Parameter Name	Usage	Data Value
1	ControlDTCSetting Request Service Id	M	0x85
subFunction = []			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [DTCSettingType]	M	0x00 - FF
	on	M	0x01 /81
	off	M	0x02 /82

Table 25: Control DTC setting

6.19 0x22 ReadDataByIdentifier

The client requests to read the current value of a record identified by a provided data identifier

ReadDataByIdentifier Request Service Id0x22			
Data Byte No	Parameter Name	Usage	Data Value
1	Read Data By Identifier Request Service Id	M	0x22
dataIdentifier[] #1 = []			
Data Byte No	Parameter Name	Usage	Data Value
2	dataIdentifier[] #1 = [byte#1 (MSB)	M	0x00 - FF
3	byte#2]	M	0x00 - FF
:	:	:	:
n-1	dataIdentifier[] #m = [byte#1 (MSB)	U	0x00 - FF
n	byte#2]	U	0x00 - FF

Table 26: Read Data By ID

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6.20 0x23 ReadMemoryByAddress

The client requests to read the current value of the provided memory range.

ReadMemoryByAddress Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	Read memory By address Request Service Id	U	0x23
AddressAndLengthFormatIdentifier			
Data Byte No	Parameter Name	Usage	Data Value
2	AddressAndLengthFormatIdentifier	M	0x00 - FF
MemoryAddress			
Data Byte No	Parameter Name	Usage	Data Value
3	MemoryAddress[] = [byte#1 (MSB)	M	0x00 - FF
:	:	:	:
6	byte#4]	M	0x00 - FF
MemorySize			
Data Byte No	Parameter Name	Usage	Data Value
7	MemorySize[] = [byte#1 (MSB)	M	0x00 - FF
8	Byte#2	M	0x00 - FF

Table 27: Read Memory by Address

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6.21 0x2A ReadDataByPeriodicIdentifier

The client requests to schedule data in the server for periodic transmission.

ReadDataByPeriodicIdentifier Request Service Id0x2A			
Data Byte No	Parameter Name	Usage	Data Value
1	Read Data By Periodic Identifier Request Service Id	U	0x2A
TransmissionMode[]			
Data Byte No	Parameter Name	Usage	Data Value
2	transmissionMode[]=[byte#1]	U	0x00 - FF
	sendAtSlowRate	U	0x01
	sendAtMediumRate	U	0x02
	sendAtFastRate	U	0x03
	stopSending	U	0x04
PeriodicDataIdentifier[]			
Data Byte No	Parameter Name	Usage	Data Value
3	PeriodicDataIdentifier[]#1	C	0x00-FF
:	:	:	:
#m+2	PeriodicDataIdentifier[]#m	U	0x00-FF

Table 28 Read data by periodic identifier

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6.22 0x2C DynamicallyDefineDataIdentifier

The client requests to dynamically define data Identifiers that may subsequently be read by the readDataByIdentifier service.

DynamicallyDefinitionDataIdentifier Request Service Id0x2C			
Data Byte No	Parameter Name	Usage	Data Value
1	DynamicallyDefinitionDataIdentifier Request Service Id	U	0x2C
Definition type[]			
Data Byte No	Parameter Name	Usage	Data Value
2	definitionType[]=[defined by identifier]	U	0x01
dynamicallyDefinedDataIdentifier[]			
Data Byte No	Parameter Name	Usage	Data Value
3	dynamicallyDefinedDataIdentifier []#1	U	0x00-FF
4	dynamicallyDefinedDataIdentifier []#2	U	0x00-FF
sourceDataIdentifier			
Data Byte No	Parameter Name	Usage	Data Value
5	sourceDataIdentifier []#1	U	0x00-FF
6	sourceDataIdentifier []#2	U	0x00-FF
PositionInSourceDataRecord			
Data Byte No	Parameter Name	Usage	Data Value
7	PositionInSourceDataRecord	U	0x00-FF
MemorySize			
Data Byte No	Parameter Name	Usage	Data Value
8	memorySize [#1]	U	0x00-FF
:	:	:	:
sourceDataIdentifier			
Data Byte No	Parameter Name	Usage	Data Value
n-3	sourceDataIdentifier []#1	U	0x00-FF
n-2	sourceDataIdentifier []#2	U	0x00-FF
PositionInSourceDataRecord			
Data Byte No	Parameter Name	Usage	Data Value

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n-1	PositionInSourceDataRecord	U	0x00-FF
MemorySize			
Data Byte No	Parameter Name	Usage	Data Value
n	memorySize [#1]	U	0x00-FF

Table 29: Dynamic defined Identifier

6.23 0x2E WriteDataByIdentifier

The client requests to write a record specified by a provided dataIdentifier.

WriteDataByIdentifier Request Service Id 0x2E			
Data Byte No	Parameter Name	Usage	Data Value
1	WriteDataByIdentifier Request Service Id	M	0x2E
dataIdentifier[]			
Data Byte No	Parameter Name	Usage	Data Value
2	dataIdentifier[] = [byte#1 (MSB)	M	0x00 - FF
3	byte#2]	M	0x00 - FF
dataRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
4	dataRecord[] = [data#1	M	0x00 - FF
:	:	:	:
m+3	data#m]	U	0x00 - FF

Table 30: Write Data by ID

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6.24 0x3D WriteMemoryByAddress

The client requests to overwrite a provided memory range.

WriteMemoryByAddress Request Service Id0x3D			
Data Byte No	Parameter Name	Usage	Data Value
1	write memory By address Request Service Id	U	0x3D
AddressAndLengthFormatIdentifier			
Data Byte No	Parameter Name	Usage	Data Value
2	AddressAndLengthFormatIdentifier	M	0x00 - FF
MemoryAddress			
Data Byte No	Parameter Name	Usage	Data Value
3	MemoryAddress[] = [byte#1 (MSB)	M	0x00 - FF
:	:	:	:
6	byte#4]	M	0x00 - FF
MemorySize			
Data Byte No	Parameter Name	Usage	Data Value
7	MemorySize[] = [byte#1 (MSB)	M	0x00 - FF
8	Byte#2	M	0x00 - FF
dataRecord			
Data Byte No	Parameter Name	Usage	Data Value
9	dataRecord[]=[data byte#1	M	0x00 – FF
:	:	:	:
#m+9	byte#m	U	0x00 – FF

Table 31: Write Memory by Address

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6.25 0x14 ClearDiagnosticInformation.

Allows the client to clear diagnostic information from the server (including DTCs, captured data, etc.)

Clear Diagnostic Request Service Id0x14			
Data Byte No	Parameter Name	Usage	Data Value
1	ClearDiagnosticInformation Request Service Id	M	0x14
groupOfDTC[]			
Data Byte No	Parameter Name	Usage	Data Value
2	groupOfDTC[] = [groupOfDTCHighByte	M	0xFF
3	groupOfDTCMiddleByte	M	0xFF
4	groupOfDTCLowByte]	M	0xFF

Table 32: Clear DTC

6.26 0x19 ReadDTCInformation

Allows the client to request diagnostic information from the server (including DTCs, captured data, etc.)

Subfunctions of Service0x19:

Sub Service	Subservice Name	Usage
0x01	reportNumberOfDTCByStatusMask	M
0x02	reportDTCByStatusMask	M
0x03	reportDTCSnapshotIdentification	U
0x04	reportDTCSnapshotRecordByDTCNumber	M
0x05	reportDTCSnapshotRecordByRecordNumber	U
0x06	reportDTCExtendedDataRecordByDTCNumber	M
0x07	reportNumberOfDTCBySeverityMask	U
0x08	reportDTCBySeverityMaskRecord	U
0x09	reportSeverityInformationOfDTC	U
0x0A	reportSupportedDTC	M
0x0B	reportFirstTestFailedDTC	M
0x0C	reportFirstConfirmedDTC	U
0x0D	reportMostRecentTestFailedDTC	U
0x0E	reportMostRecentConfirmedDTC	U
0x0F	reportMirrorMemoryDTCByStatusMask	U
0x10	reportMirrorMemoryDTCExtendedDataRecordByDTCNumber	U
0x11	reportNumberOfMirrorMemoryDTCByStatusMask	U
0x12	reportNumberOfEmissionsRelatedOBDDTCByStatusMask	U

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0x13	reportEmissionsRelatedOBDDTCByStatusMask	U
0x14	reportDTCFaultDetectionCounter	U
0x15	reportDTCWithPermanentStatus	U

Table 33: Subfunction Service0x19

6.27 0x2F InputOutputControlByIdentifier

The client requests the control of an input/output specific to the server. For full implementation of enabling mask please see ISO14229-1(2020) table 400 section 13.2.2.3

InputOutputControlByIdentifier Request Service Id0x2F			
Data Byte No	Parameter Name	Usage	Data Value
1	InputOutputControlByIdentifier Request Service Id	M	0x2F
dataIdentifier []			
Data Byte No	Parameter Name	Usage	Data Value
2	dataIdentifier#1[] = [byte#1 (MSB)	M	0x00 - FF
3	byte#2 (LSB)]	M	0x00 - FF
controlOptionRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
4	controlOptionRecord#1[] = [controlState#1/inputOutputControlParameter	M	0x00 - FF
:	:	:	:
4+(m-1)	controlState#m]	C	0x00 - FF
controlEnableMaskRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
4+m	controlEnableMaskRecord#1[] = [controlMask#1	C	0x00 - FF
:	:	:	:
4+m+(r-1)	controlMask#r]	C	0x00 - FF

For the inputOutputControlParameter, following parameters defined:

Hex	Description	Cvt
0x00	returnControlToECU	M
0x01	resetToDefault	U
0x02	freezeCurrentState	U
0x03	shortTermAdjustment	M

Table: 34 I/O Control by ID

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6.28 0x31 Routine Control

The client requests to start, stop a routine in the server(s) or requests the routine results. All routines are required to run in a non-default session i.e., Extended and programming sessions based on their use case. Routines are not allowed to run in default session when they are called externally via a tester.

RoutineControl Request Service Id 0x31			
Data Byte No	Parameter Name	Usage	Data Value
1	RoutineControl Request Service Id	M	0x31
sub-function = [routineControlType]			
Data Byte No	Parameter Name	Usage	Data Value
2	startRoutine	M	0x01
2	stopRoutine	U	0x02
2	requestRoutineResults	U	0x03
routineIdentifier []			
Data Byte No	Parameter Name	Usage	Data Value
3	routineIdentifier [] = [byte#1 (MSB)	M	0x00 - FF
4	byte#2]	M	0x00 - FF
routineControlOptionRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
5	routineControlOptionRecord[] = [routineControlOption#1	C/U	0x00 - FF
:	:	:	:
n	routineControlOption#m]	C/U	0x00 - FF

Table: 35 Routine Control

6.29 0x34 RequestDownload

The client requests the negotiation of a data transfer from the client to the server.

6.30 0x35 RequestUpload

The client requests the negotiation of a data transfer from the server to the client. This service shall not be supported due security concerns.

6.31 0x36 TransferData

The client transmits data to the server (download) or requests data from the server (upload).

6.32 0x37 RequestTransferExit

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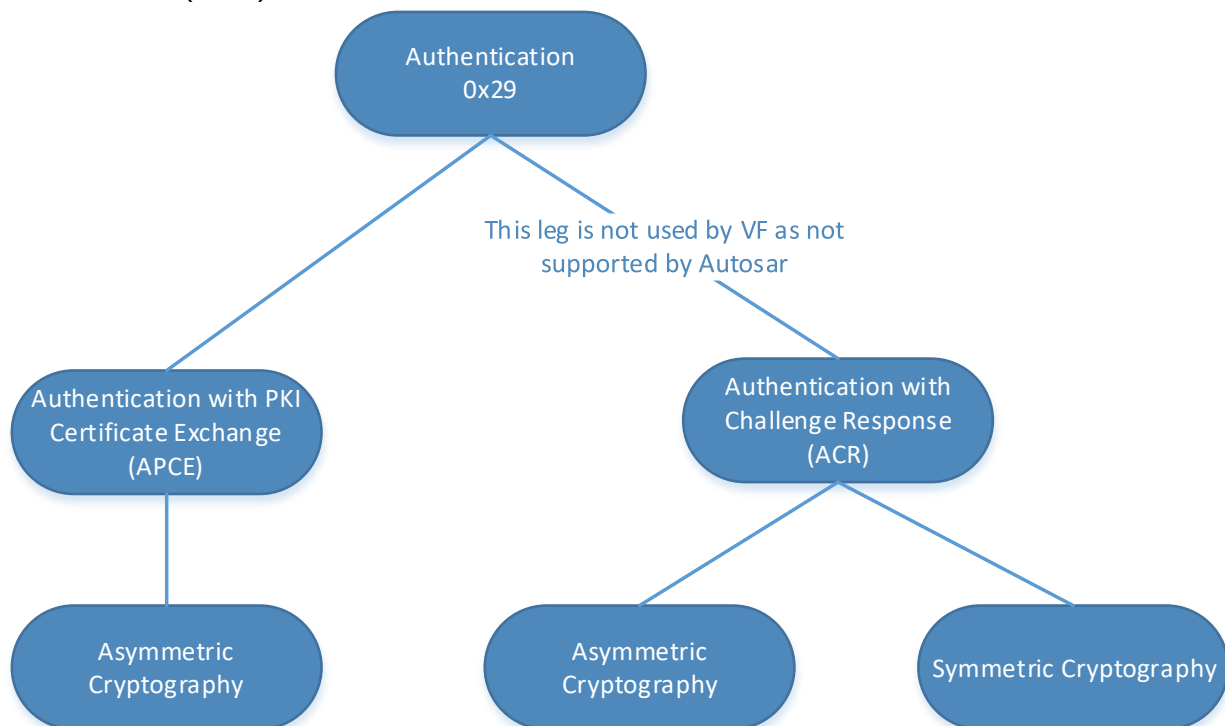
VFDSVAEEP0001-V1.6

The client requests the termination of a data transfer.

Note: - that all above services' definition must be compliant with ISO 14229 standard, including diagnostic request, positive response, all negative responses, and negative response code (NRC) shall also be implemented in accordance with current definition of ISO requirements. Any deviation must be aligned with OEM diagnostic team prior to implementing these in software.

6.33 0x29 Authentication

For details on the description of how this service is fully implemented and used please refer to ISO14229-1 (2020) or latest

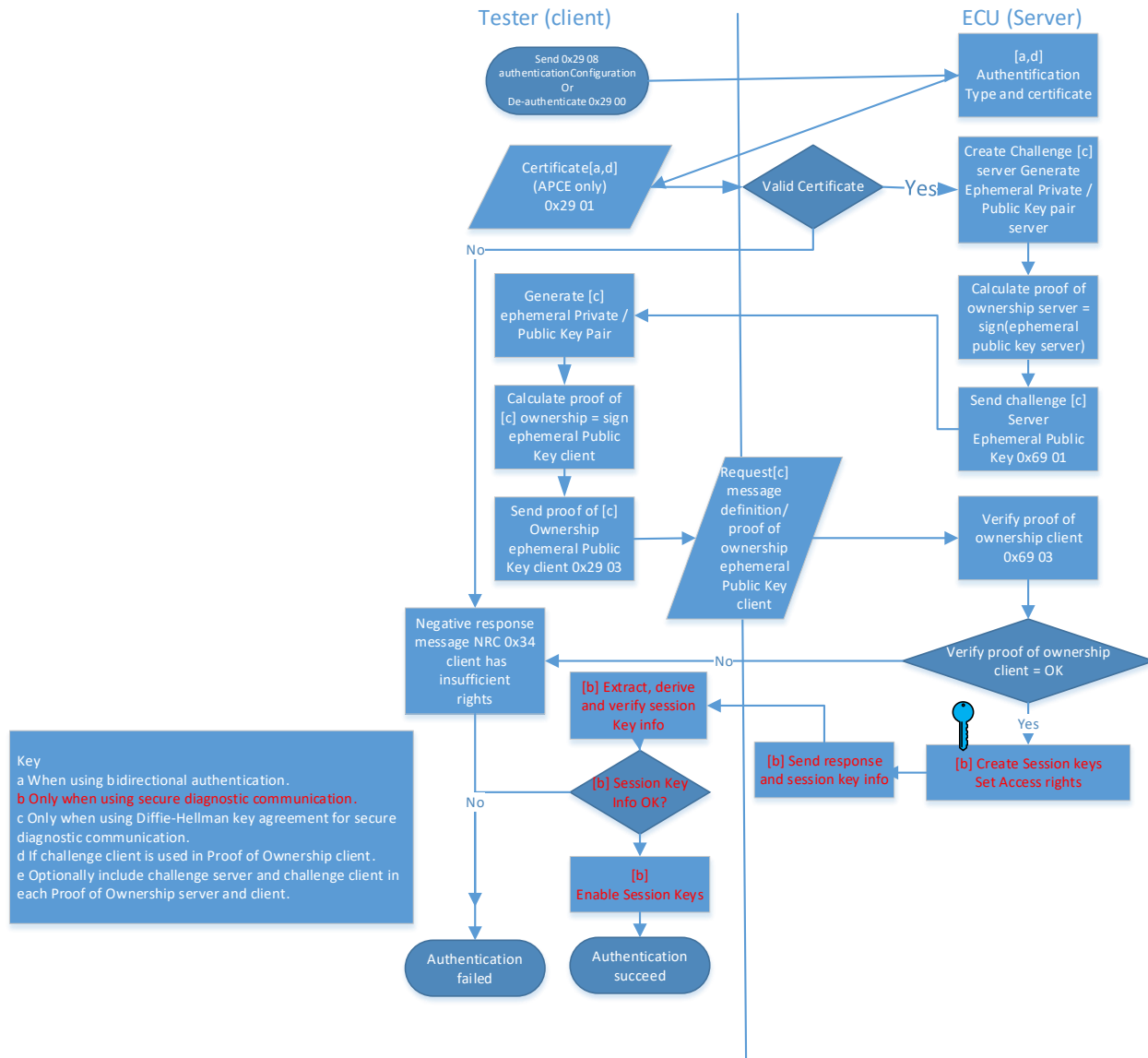


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Authentication0x29			
Data Byte No	Parameter Name	Usage	Data Value
1	Authentication Request SID	M	0x29
sub-function			
Data Byte No	Parameter Name	Usage	Data Value
2	DeAuthenticate (request to leave Authenticated state)	M	0x00
2	Verify Certificate Unidirectional (Initiate Authentication by verifying the Certificate).	M	0x01
2	Verify Certificate Bidirectional (Initiate Authentication by verifying the Certificate and generating a Proof of Ownership from the server).	M	0x02
2	Proof Of Ownership (Verify the Proof of Ownership from the client).	M	0x03
2	Transmit Certificate Verify Certificate and extract information from Certificate to handle it according to its contents.	M	0x04
2	Authentication Configuration Indicates the provided authentication configuration	M	0x08

6.34 Service 0x29 application flow



6.35 DIDs

For processes in Production and/or in Service it is necessary to use a number of common basic DIDs, which will be used for identification of components, software or the complete vehicle.

The number and content of this common DIDs must be clarified together with the Logistic Team, Production and Service and should have the same structure for all used ECUs. Deviations must be agreed with OEM diagnostic team.

A minimum of common Basic DIDs are defined in the document:

- Basic_DID_Definition

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6.36 ECU Specific DIDs

All ECU specific Data/Parameter Identifier must be described in the respective component specification. And OEM ECU engineer and ECU supplier are responsible for the definition of DIDs' parameter, content, formula, format, unit and using method and details of its implementation shall be described in the ODX template to assist manufacturing Aftersales teams.

Any DTCs which is triggered by specific event such as a measured voltage or a specific value of a signal shall support a DID indicating the measured value being reported.

6.37 Supplier Diagnostic Quality Report

The Supplier of the ECU must deliver a Diagnostic Quality Report. The content and design of the report is defined by the Diagnostic Department and should not be changed. The Supplier should update the Report for any Function Release Step and send the full filled report to the Integration team during the relevant Function Delivery step. The content of the Diagnostic Quality Report must be filled completely by the Supplier. The Supplier must fill the report truly and accurately.

7 Guidelines for ECU Diagnostics

7.1 Message to ECU Suppliers

- Maximize your ECU diagnostics to minimize the no fault found returns (warranty parts)
 - This applies to DTC's, DID's, RID's etc
- All I/O circuits shall be diagnosable against short to battery/ground/open circuit where possible. This will be dependent on ECU design
- All inputs (including Sensors) shall be supported by Service0x22 and the input readable via DID where possible.
- All outputs shall be supported by Service 0x2F or where possible by Service 0x31 to ensure correct usage of the output function
- ECU shall have a DTC for all unexpected vehicle behavior caused by
- Plausibility/Rationality faults
- Loss of Communications
- Invalid Data Received from another ECU
- Checksum and Alive Counter Errors detected in received CAN messages
- Where ECU software determines it has not correctly performed one of its functions, or otherwise decides upon a vehicle reaction that a customer may notice.

7.2 Assigning DTC numbers

- DTC's are assigned to an ECU per ISO-15031-6 / SAE-J2012DA, else for Manufacturer Controlled Ranges please follow VINFAST DTC Ranges spreadsheet (see below).
- Between VINFAST EE Diagnostic Group and ECU Supplier, aim to commonize DTC assignments across all ECU's.

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Note reference Documents shall be checked for any updates or clarifications when using the content contained in sections 7.2 through to 7.4

1. ECU DTC

System	System DTC	ECU	DTC number range of DTC high&middle byte ISO15031-6 (and J2012-DA) - ISO defined	DTC number range of DTC high&middle byte ISO15031-6 Manufacturer controlled
Chassis	Cxxxx	EPS	Please refer to SAE J2012-DA.	C1300-C13FF / 0x5300-0x53FF
		ESC		C1500-C15FF / 0x5500-0x55FF
		VCU		P1000-P10FF / 0x1000-0x10FF
		BMS		P1200-P12FF / 0x1200-0x12FF
		EDS-F & EDS-R		P1800-P18FF / 0x1800-0x18FF
		3in1(PDM)		P1400-P14FF / 0x1400-0x14FF
		AVAS		P1600-P167F / 0x1600-0x167F
		EAC (BEV Plus)		P1680-P16FF / 0x1680-0x16FF
		GS		P1500-P157F / 0x1500-0x157F
		TCU		P1580-P15FF / 0x1580-0x15FF
Powertrain	Pxxxx	EMS		P1700-P17FF / 0x1700-0x17FF
				P1180-P11FF / 0x1180-0x11FF
				P1100-P117F / 0x1100-0x117F
		ACM		B1400-B14FF / 0x3400-0x34FF
		ADAS		B1900-B19FF / 0x3900-0x39FF
		APM		B1100-B117F / 0x3100-0x317F
		BCM		B1000-B10FF / 0x3000-0x30FF
		CCM		B1700-B17FF / 0x3700-0x37FF
		CPD		B1180-B11FF / 0x3180-0x31FF
		ETG		B1200-B127F / 0x3200-0x327F
Body	Bxxxx	HUD		B1280-B12FF / 0x3280-0x32FF
		IPC		B1800-B18FF / 0x3800-0x38FF
		MFS		B1500-B15FF / 0x3500-0x35FF
		MHU		B1600-B16FF / 0x3600-0x36FF
		OCS		B1300-B137F / 0x3300-0x337F
		PSM-D		B1380-B13FF / 0x3380-0x33FF
		SHVU-F		B2100-B217F / 0x4100-0x417F
		SHVU-R		B2180-B21FF / 0x4180-0x41FF
		TRM		B2000-B20FF / 0x4000-0x40FF
		XGW		
		Network	Uxxxx	ALL ECU

Notes:
Step 1. DTC shall first be assigned according to the trouble code defined in ISO15031-6 (and J2012-DA).
Step 2. If the corresponding trouble code cannot be found, the code shall be selected within the scope defined by Manufacturer controlled range in column E.

Figure 8. Screenshot from VF35 DTC Ranges Spreadsheet

CAN Bus	LIN Master Module	LIN 1	LIN 2	LIN 3	LIN 4	LIN 5
Body	APM	U1200-U121F				
Body	BCM	U1220-U122F	U1230-U123F	U1240-U124F		
Info	CCU	U1250-U125F				
Info	HCM	U1260-U126F				
PTCAN	ADAS	U1270-U127F	U1280-U128F			
PTCAN	EMS	U1290-U129F				
PTCAN	VCU	U12A0-U12AF				
AllCAN	XGW	U12B0-U12BF	U12C0-U12CF			

Figure 9. Screenshot from VF35 DTC Ranges Spreadsheet for LIN Master DTC Ranges

7.3 General Rules for U-Codes

- At a minimum, the ECU shall supervise the fastest periodic highest priority CAN message received from another ECU according to the CAN matrix.
- For Loss Of Communication DTC's, the Rx ECU can only report a DTC against ECU's that reside on the same CAN network.
- In the case where the Rx CAN message originates from an ECU on another CAN network, the Rx ECU shall report a Loss Of Communications with the Gateway (XGW), and the Gateway

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(XGW) shall report the Loss Of Communications DTC against the Tx ECU on the other CAN bus

- The Receiving ECU must supervise at least 1 message from each source ECU (OR'd) in its DTC Test Failed logic, and with individual message timeouts.
See below example

DTC Number (hex)	DTC Number	Failure Type Byte and meaning	DTC Name	DTC Failure Type Definition (if necessary)	Repair Action	Test Failed criteria/ Confirmation Failure criteria	Mature Time	Test Pass Criteria
C14687	U014687	87 - Missing Message	Lost Communication With Serial Data Gateway Module "A" - Missing Message	Missing message GW_FOTA_StaConfirm_0x3c, or GW_MPS_Control_Button_0x3f7, or GW_IPC_Obstacle_0x3f8, or GW_EPS_Status_0x1f0, or XGW_Timestamp_0x2f2, or GW_BMS_Sta_0x215, or GW_MHU_FOTA_STATUS_0x332	Check XGW for DTC's first since Source ECU is likely on another Communication Network	10message cycle times; 10 periodic according to CAN matrix; no message received (11000150000 ms)	0x3FC (5s) 0x3F7 (10s) 0x3F8 (10s) 0x1F0 (200ms) 0x2F2 (1000ms) 0x215 (500ms) 0x332 (1000ms)	The message frame is received

Figure 10. Example of U0146-87 DTC Failure Type Definition for multiple message supervision

- If a Rx CAN message has an Alive Counter and Checksum signals, this allows End To End Protection of that message from the source to the destination ECU's, even if they reside on different CAN networks.
- When an Alive Counter or Checksum failure is detected, the Rx ECU shall report the DTC in the following form (Uxxxx.YY, where YY is either FTB=0x82 or 0x83)
- For Rx signals with states 0x1=Invalid/0x00=Valid or 0x00=Valid/0x01=Invalid (see Figure 4 example below from Info_CAN), if signal is received with "Invalid" state, the Rx ECU shall stop using the payload signal (since no longer valid), and report DTC UxxxxYY (where YY is FTB=81). In the case of VehicleSpdSts = 0x01 (Invalid) all Receivers listed shall report DTC Uxxxx81 (Invalid Data Received From ECU xxxx) and stop using payload signal VehicleSpd until VehicleSpdSts = 0x00 (Valid)
 - Transmitter Module must also report a DTC to identify the root cause of the "Invalid" state

Bus	Signal Name	Message Name	Message ID (hex)	Value Matrix	Sender	Receiver
Info-CAN	ADAS_TSR_TSSpd1	ADAS_TSR	0x129	n.a.	ADAS_Info	MHU, HUD
Info-CAN	ADAS_TSR_TSSpdVal1	ADAS_TSR	0x129	0 "Valid" 1 "Invalid"	ADAS_Info	MHU, HUD
Info-CAN	ADAS_TSR_TSSpd2	ADAS_TSR	0x129	n.a.	ADAS_Info	MHU, HUD
Info-CAN	ADAS_TSR_TSSpdVal2	ADAS_TSR	0x129	0 "Valid" 1 "Invalid"	ADAS_Info	MHU, HUD
Info-CAN	ADAS_TSR_TSSpd3	ADAS_TSR	0x129	n.a.	ADAS_Info	MHU, HUD
Info-CAN	ADAS_TSR_TSSpdVal3	ADAS_TSR	0x129	0 "Valid" 1 "Invalid"	ADAS_Info	MHU, HUD
Info-CAN	MCPPressure	ESC_STATUS	0x20D	n.a.	XGW_Info	MHU, HUD
Info-CAN	MCPPressureSts	ESC_STATUS	0x20D	1 "Invalid" 0 "Valid"	XGW_Info	MHU, HUD
Info-CAN	SAS_SteerWheelAngle	0x17E	n.a.	n.a.	XGW_Info	MHU
Info-CAN	SAS_SteeringAngleValid	SAS_Sensor	0x17E	1 "Valid" 0 "Invalid"	XGW_Info	MHU
Info-CAN	SAS_SteerWhlRotSpd	SAS_Sensor	0x17E	n.a.	XGW_Info	MHU
Info-CAN	SAS_SteerWhlRotSpdStatus	SAS_Sensor	0x17E	1 "Valid" 0 "Invalid"	XGW_Info	MHU
Info-CAN	VehicleSpd	ESC_STATUS	0x20D	n.a.	XGW_Info	MHU, HUD, CPD
Info-CAN	VehicleSpdSts	ESC_STATUS	0x20D	1 "Invalid" 0 "Valid"	XGW_Info	MHU, HUD, CPD

Figure 11. Extract from VF35 CAN Matrix (Info-CAN)

- Where ECU is receiving multiple messages with Rolling Count/Checksum or Validity signals, the DTC failure criteria shall "OR" the messages into a signal DTC. No need to create independent DTC's for each message failure. See examples below

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DTC Number (hex)	DTC Number	Failure Type Byte and meaning	DTC Name	DTC Failure Type Definition(if necessary)	Repair Action	Test Failed criteria/ Confirmation Failure criteria	Mature Time	Test Pass Criteria
C41581	U041581	01 - Invalid Serial Data Received	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Invalid Serial Data Received	ESC message received with INVALID signal state	Check the relevant ECU for performance DTC	ESC message 0x332 is received with signal ABS_LFVWhSpdRawValid = \$0 (Invalid), or ABS_RFVWhSpdRawValid = \$0 (Invalid), or ESC message 0x342 is received with signal ABS_LRWHSpdRawValid = \$0 (Invalid), or ABS_SRWHSpdRawValid = \$0 (Invalid), or ABS_VehSpdValid = \$0 (Invalid), or ESC message 0x3E2 is received with signal ESC_LateralACC_Valid = \$0 (Invalid), or ESC_LongitudeACC_Valid = \$0 (Invalid), or ESC_VehDynYawRate_Valid = \$0 (Invalid)	Immediate	ESC message 0x332 is received with signal ABS_LFVWhSpdRawValid = \$1 (Valid), and ABS_RFVWhSpdRawValid = \$1 (Valid), and ESC message 0x342 is received with signal ABS_LRWHSpdRawValid = \$1 (Valid), and ABS_SRWHSpdRawValid = \$1 (Valid), and ABS_VehSpdValid = \$1 (Valid), and ESC message 0x3E2 is received with signal ESC_LateralACC_Valid = \$1 (Valid), and ESC_LongitudeACC_Valid = \$1 (Valid), and ESC_VehDynYawRate_Valid = \$1 (Valid)
C45782	U045782	02 - Alive / Sequence Counter Incorrect / Not Updated	Invalid Data Received From Information Center "A" - Alive / Sequence Counter Incorrect / Not Updated	Message received from MHU without the corresponding rolling count value being properly updated.	Check the relevant ECUs	Alive counter is not updated for message MHU_STATUS_0x422, or MHU_FOTA_STATUS_0x432	Immediate	Alive counter is properly updated for message MHU_STATUS_0x422, and MHU_FOTA_STATUS_0x432
C45783	U045783	03 - Value of Signal Protection Calculation Incorrect	Invalid Data Received From Information Center "A" - Value of Signal Protection Calculation Incorrect	Message received from MHU with an incorrect checksum calculation.	Check the relevant ECUs	Checksum of message MHU_STATUS_0x422, or MHU_FOTA_STATUS_0x432 is incorrect.	Immediate	Checksum of message MHU_STATUS_0x422, and MHU_FOTA_STATUS_0x432 is correctly received

Figure 12. Example strategy for Invalid Data from MHU, monitoring multiple messages same source

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- When Assigning Loss Of Communication/Invalid Data Received from ECU "XXX"/Bus OFF DTC's shall refer to figure 6 so consistency can be maintained for all ECU's
- ECU's shall support Bus OFF DTC for each CAN network they are connected too.

VF module name	SAE J2012 module name	Lost comms DT	Invalid data DT	Comment
APM	Anti-Pinch Module	U1034	U160A	
CCU	Climate Control Unit	U1039	U0424	
PSDM-D	"Seat Control Module A"	U0208	U0509	
BCM	Body Control Module	U0140	U0422	
ACM / ACU	Airbag Control Module	U1038	U0452	Vf35 = ACM, Vf34 = ACU
ETG	Rear Gate Module	U0230	U0531	
TRM	Trailer Module	U103C	U160C	
SHVU-F	"Seat Control Module C"	U0210	U0511	Reserving "seat control module b" for passenger seat memory module
SHVU-R	"Seat Control Module D"	U0211	U0522	
OCS	Restraints Occupant Classification System	U0154	U0455	
IPC	Instrument Panel Cluster (IPC) Control Module	U0155	U0423	
XGW	Serial Data Gateway Module "A"	U0146	U0447	
BMS	Battery Energy Control Module "A"	U0111	U0412	
AVAS	Audible Alert Control Module	U0160	U0461	
OBC	Battery Charger Control Module "A"	U019B	U049C	
DCDC	DC/DC Converter Control Module "A"	U0298	U0539	
VCU	Vehicle Dynamics Control Module	U0122	U0416	Not using "Hybrid/EV Powertrain Control Module" (U0293/U0534)
EDS-F	Drive Motor Control Module A	U0110	U0411	
EDS-R	Drive Motor Control Module B	U0292	U0593	
ESC	Anti-Lock Brake System (ABS) Control Module	U0121	U0415	
EPS	Power Steering Control Module "A"	U0131	U0420	
ADAS	Automated Driving System Control Module "A"	U015C	U045D	
CPD	Child Presence Detection Module	U103B	U160B	
HUD	Head Up Display	U0158	U0459	
MHU	Media Head Unit	U0156	U0457	
MDU	Media Display Unit	U0257	U0558	
EMS	ECM/PCM "A"	U0100	U0401	
TCU	TCM	U0101	U0402	
GS / EKS	Transmission Range Control Module	U019E	U049F	Vf35 = GS, Vf34 = EKS
RCU	Redundant Control Unit	U110D	U160D	
IDB	Intelligent Distribution Brakes	U110E	U160E	
FDU	Power Distribution Unit + OBC + DCDC	U110F	U160F	Vf35 only
MFS	Multi Function Switches	U1150	U1630	
DWS	Driver Window Switch	U1221		
LI2-SPGM	Sunroof Control Module	U0163		
LI1-BAS	Battery Monitor Module	U01B0		
LI2-WLC	Wireless Accessory Charging Module "A"	U020C		
LI1-SCM	Steering Column Control Module	U0212		
LI1-RLS	Rain Sensing Module	U0231		
LI2-SRBM	Sunroof Blind Module	U1831		
LI2-FKS	Foot Kick Sensor	U1832		
LI3-ALM1	LI3 ambient lighting module 1	U1841		
LI3-ALM2	LI3 ambient lighting module 2	U1842		
LI3-ALM3	LI3 ambient lighting module 3	U1843		
LI3-ALM4	LI3 ambient lighting module 4	U1844		
LI3-ALM5	LI3 ambient lighting module 5	U1845		
LI3-ALM6	LI3 ambient lighting module 6	U1846		
Bus OFF DTC's	DTC Name			
	Chassis CAN	U11T1		Failure Type Byte 88
	Powertrain CAN	U11T2		Failure Type Byte 88
	Info CAN	U11T3		Failure Type Byte 88
	Body CAN	U11T4		Failure Type Byte 88
	Diag CAN	U11T5		Failure Type Byte 88
	Flexray Bus Fault	U11T6		Failure Type Byte 88
	FA-CAN	U11T7		Failure Type Byte 88
	BA-CAN	U11T8		Failure Type Byte 88
Battery High/Low DT	DTC Name			
	Power Supply - Circuit Voltage Above Threshold	U1100		Failure Type Byte 17
	Power Supply - Circuit Voltage Below Threshold	U1101		Failure Type Byte 16
Coding DTC's	DTC Name			
	Coding Data String (CDS) - Not Configured	U1500		Failure Type Byte 55 (Not Configured) or 56 (Invalid / Incompatible Configuration)
	Factory Vehicle Order Data (VOD)	U1501		Failure Type Byte 55 (Not Configured) or 56 (Invalid / Incompatible Configuration)
	Current Vehicle Order Data (VOD)	U1502		
	Current Vehicle Order Data (VOD) Mismatch with VOD Master	U1503		
VIN DTC's	DTC Name			
	Vehicle Identification Number - Not Configured	U3002		Failure Type Byte 55 (Not Configured)
Yellow highlighting indicates DTC not aligned to SAE J2012				

Figure 13. Communication DTC list

Contact EE-Diagnostic Engineer, or Aftersales Diagnostic and Programming team if you require DTC not listed above

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7.4 DTC Names

- If using a DTC Number from the J2012DA list, then the DTC name shall match J2012DA
- If defining a DTC from the Manufacturer Controlled Range, please ensure the DTC name is intuitive
 - Please keep the customer in mind (the Service Technician)
 - We should avoid acronym's in DTC Name unless they are obvious industry standards
 - Refer to the Vinfast Diagnostics team for guidance where unsure
- Where defining a single DTC with multiple FTB's, the DTC name shall be same (see example in Figure 7 below). The FTB will differentiate the failure mode of the DTC

DTC Number (hex)	DTC Number	Failure Type Byte and meaning	DTC Name
970D15	B170D15	0x15: Circuit short to battery or open	Air Distribution Feedback Signal
970D11	B170D11	0x11: Circuit short to ground	Air Distribution Feedback Signal

Figure 7. See DTC B170D, with unique FTB

7.5 Configuration Related DTC's to consider (examples)

- VIN not Written or VIN Invalid
- Coding not written or Coding Invalid
- Key Fobs not Programmed
- Immobilizer Key Not Learned
- Steering Angle Sensor not Learned
- Airbag Module not Armed
- ADAS Camera/Radars not learned

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7.6 Assigning DID Numbers

For ECU DID's (those not defined in the Basic DID Definition Specification), the ISO-14229 Annex C table C.1 shall be referenced for permitted ranges

7.7 Supplier DID's shall be defined from System Supplier Specific Ranges below.

Supplier DID Range	Comments
0xFD00-0xFEFF	This range of values shall be used to reference system supplier specific record data identifiers and input/output identifiers within the server.

Figure 8. System Supplier Specific DID Range

7.8 Assigning RID Numbers

- For ECU specific RID's (those not defined in VINFAST Specifications), ISO-14229 Annex F table F.1 shall be referenced for permitted ranges
- Typically Supplier RID's will be defined from *System Supplier Specific Range* below

0x0200 - 0xDFFF	vehicleManufacturerSpecific This range of values is reserved for vehicle manufacturer specific use.	U	VMS_
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Figure 9. System Supplier Specific RID Range